

FIT5217

Natural Language Processing

Week 4 – Syntactic Parsing

Trang Vu

(Slides and material built on the foundational work of : Ray Mooney and Nigel Collier, Ryan Cotterell, Julia Hockenmaier

Last Week

NLP tasks dealing with words...

- E.g. POS-tagging

... requiring finite-state representations,

- Finite-state machine

... the corresponding probabilistic models,

- Finite-state machine with probabilistic transition, Hidden Markov Models
- Learning: Maximum Likelihood Estimation (MLE)

... and appropriate search algorithms

- Dynamic programming: Viterbi

This Week

NLP tasks dealing with sentences...

- Syntactic parsing, dependency parsing

... require (at least) context-free representations,

- Context-free grammars, dependency grammars

... the corresponding probabilistic models,

- Probabilistic Context-Free Grammars (PCFG)

... and appropriate search algorithms

- Dynamic programming: CKY parsing

Overview

1. **Syntax**
2. Syntactic Parsing
3. CKY Parsing
4. Statistical Parsing
5. Probabilistic CKY Parsing
6. PCFG Training
7. Lexicalized PCFGs
8. Evaluating model performance
9. Alternatives
10. New Developments

What is the structure of a sentence?

- Sentence structure is **hierarchical**:
 - A sentence consists of **words** (I, ate, the spaghetti, with, meatballs)
 - ...which form phrases or **constituents**: “the spaghetti with meatballs”
- Sentence structure defines dependencies between words or phrases (constituents):



Phrase Chunking

- Find all non-recursive noun phrases (NPs) and verb phrases (VPs) in a sentence.
 - [NP I] [VP ate] [NP the spaghetti] [PP with] [NP meatballs].
 - [NP He] [VP reckons] [NP the current account deficit] [VP will narrow] [PP to] [NP only # 1.8 billion] [PP in] [NP September]

Phrase Chunking as Sequence Labelling

- Tag individual words with one of 3 tags
 - B (Begin) word starts new target phrase
 - I (Inside) word is part of target phrase but not the first word
 - O (Other) word is not part of target phrase
 - Sample for NP chunking
 - He reckons the current account deficit will narrow to only # 1.8 billion in September.
- Begin Inside Other

Syntax

A theory of syntax should explain which sentences are well-formed (grammatical) and which are not:

- Note that well-formed is distinct from being meaningful
- Otherwise, the mathematical study of word order



Colorless green ideas sleep furiously. ([Image source](#))

A **parser** is an algorithm for analysing a sentence given a grammar.

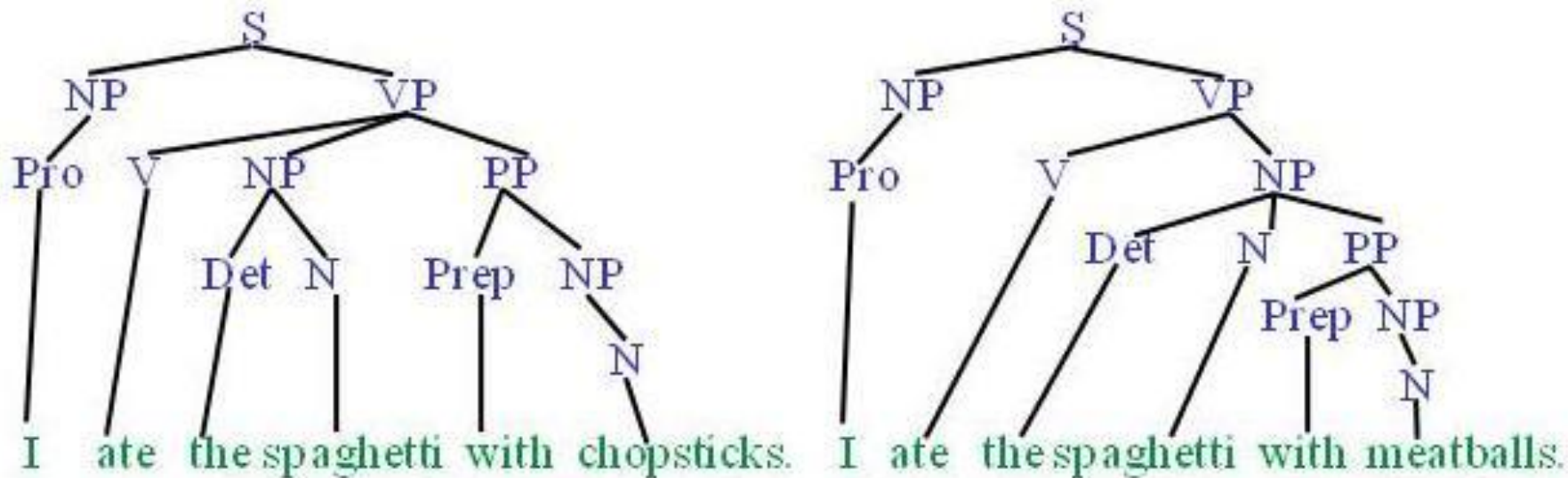
- It may just answer yes/no to answer *Does this sentence conform to the grammar?*
- It may also produce a structure description (**parse tree**) for sentences.

Overview

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2. **Syntactic Parsing**
3. CKY Parsing
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Syntactic Parsing

- Produce the correct syntactic parse tree for a sentence.



Context Free Grammars (CFG)

A context free grammar is a 4-tuple $G=(S,\Sigma,N,R)$:

- N a set of *non-terminal symbols* (or *variables*)
 - e.g. $N=\{S, NP, VP, PP, Noun, Verb, \dots\}$
- Σ a set of *terminal symbols* (disjoint from N)
 - e.g. $\Sigma = \{I, you, he, eat, drink, meatball, \dots\}$
- R a set of *productions* or *rules*
 - $R \subseteq \{A \rightarrow \beta\}$ where A is a non-terminal ($A \in N$)
and β is a string of symbols from $(\Sigma \cup N)^*$
- S , a designated non-terminal called the *start symbol*

CFG defines phrase structure trees

NP → I
NP → sushi
NP → tuna
NP → NP PP
P → with
PP → P NP
S → NP VP
V → eat
VP → V NP

NP: Noun Phrase

P: Preposition

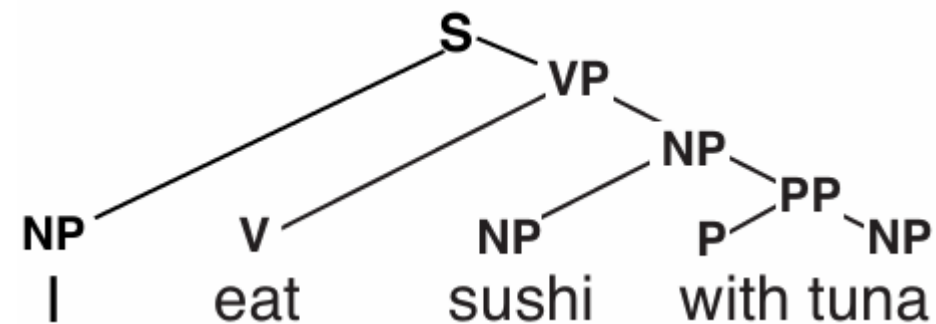
S: Sentence

PP: Prepositional Phrase

V: Verb

VP: Verb Phrase

- **Leaf nodes** (I, eat, ...) correspond to the words in the sentence
- **Intermediate nodes** (NP, VP, PP) span substrings (= the *yield* of the node), and correspond to nonterminal *constituents*
- **The root** spans the entire sentence and is labelled with the start symbol of the grammar (S)



A toy CFG

Grammar

S → **NP VP**

S → **Aux NP VP**

S → **VP**

NP → **Pronoun**

NP → **Proper-Noun**

NP → **Det Nominal**

Nominal → **Noun**

Nominal → **Nominal Noun**

Nominal → **Nominal PP**

VP → **Verb**

VP → **Verb NP**

VP → **VP PP**

PP → **Prep NP**

Lexicon

Det → **the | a | that | this**

Noun → **book | flight | meal | money**

Verb → **book | include | prefer**

Pronoun → **I | he | she | me**

Proper-Noun → **Houston | NWA**

Aux → **does**

Prep → **from | to | on | near | through**

Sentence Generation

Grammar

$S \rightarrow NP VP$

$S \rightarrow \text{Aux } NP VP$

$S \rightarrow VP$

$NP \rightarrow \text{Pronoun}$

$NP \rightarrow \text{Proper-Noun}$

$NP \rightarrow \text{Det Nominal}$

$\text{Nominal} \rightarrow \text{Noun}$

$\text{Nominal} \rightarrow \text{Nominal Noun}$

$\text{Nominal} \rightarrow \text{Nominal PP}$

$VP \rightarrow \text{Verb}$

$VP \rightarrow \text{Verb NP}$

$VP \rightarrow VP PP$

$PP \rightarrow \text{Prep NP}$

Lexicon

$\text{Det} \rightarrow \text{the} \mid \text{a} \mid \text{that} \mid \text{this}$

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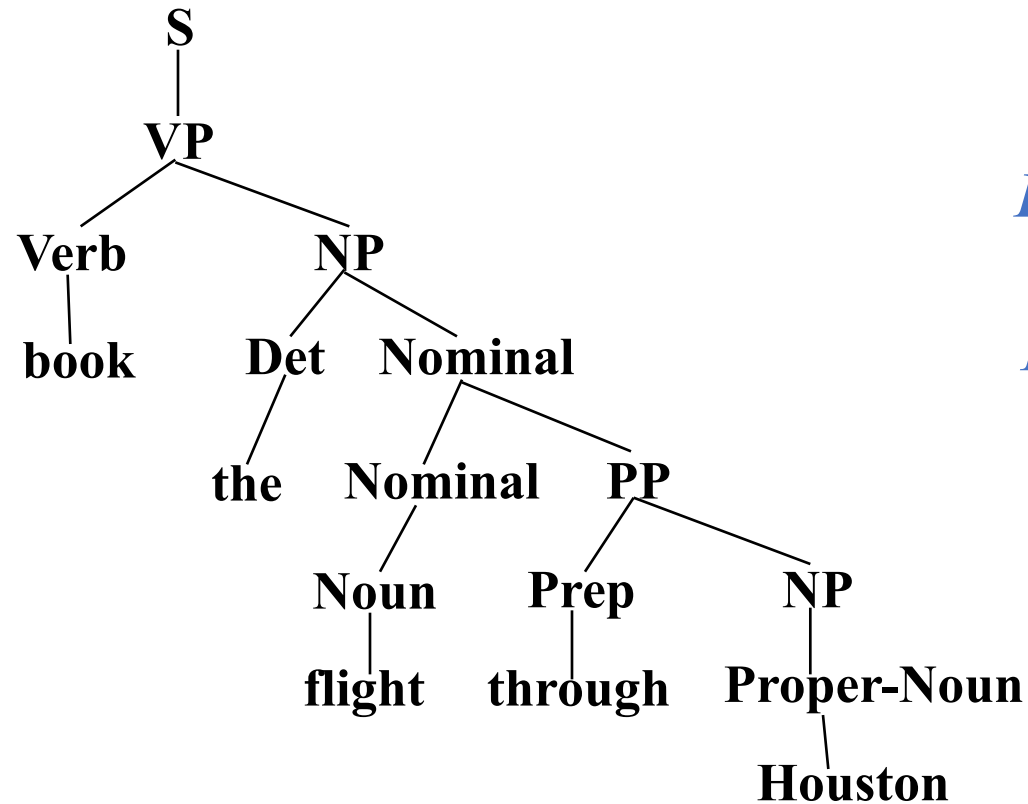
$\text{Pronoun} \rightarrow \text{I} \mid \text{he} \mid \text{she} \mid \text{me}$

$\text{Proper-Noun} \rightarrow \text{Houston} \mid \text{NWA}$

$\text{Aux} \rightarrow \text{does}$

$\text{Prep} \rightarrow \text{from} \mid \text{to} \mid \text{on} \mid \text{near} \mid \text{through}$

- Sentences are generated by recursively rewriting the start symbol using the productions until only terminal symbols remain.



*Derivation
or
Parse Tree*

Parsing

- Given a string of terminals and a CFG, determine if the string can be generated by the CFG.
 - Also return all possible parse trees for the string
- Must search the space of derivations for one that derives the given string.
 - **Top-Down Parsing**: Start searching space of derivations for the start symbol.
 - **Bottom-up Parsing**: Start search space of reverse derivations from the terminal symbols in the string.

Parsing Example

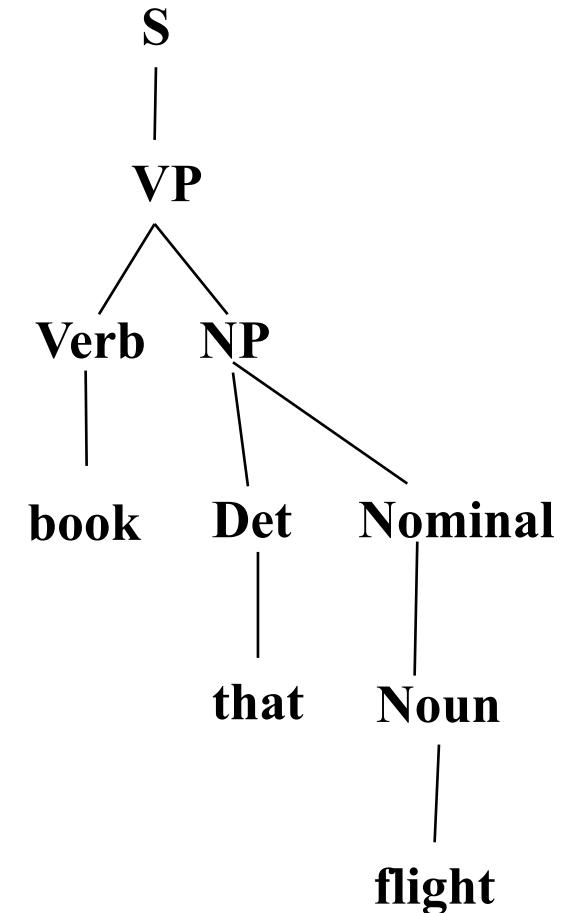
Grammar

$S \rightarrow NP VP$
 $S \rightarrow Aux NP VP$
 $S \rightarrow VP$
 $NP \rightarrow Pronoun$
 $NP \rightarrow Proper-Noun$
 $NP \rightarrow Det Nominal$
 $Nominal \rightarrow Noun$
 $Nominal \rightarrow Nominal Noun$
 $Nominal \rightarrow Nominal PP$
 $VP \rightarrow Verb$
 $VP \rightarrow Verb NP$
 $VP \rightarrow VP PP$
 $PP \rightarrow Prep NP$

Lexicon

$Det \rightarrow the \mid a \mid that \mid this$
 $Noun \rightarrow book \mid flight \mid meal \mid money$
 $Verb \rightarrow book \mid include \mid prefer$
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 $Aux \rightarrow does$
 $Prep \rightarrow from \mid to \mid on \mid near \mid through$

book that flight



Top Down Parsing

Grammar

S → NP VP

S → Aux NP VP

S → VP

NP → Pronoun

NP → Proper-Noun

NP → Det Nominal

Nominal → Noun

Nominal → Nominal Noun

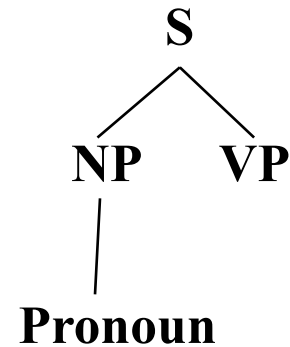
Nominal → Nominal PP

VP → Verb

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Top Down Parsing

Grammar

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$S \rightarrow VP$

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$NP \rightarrow Det Nominal$

$Nominal \rightarrow Noun$

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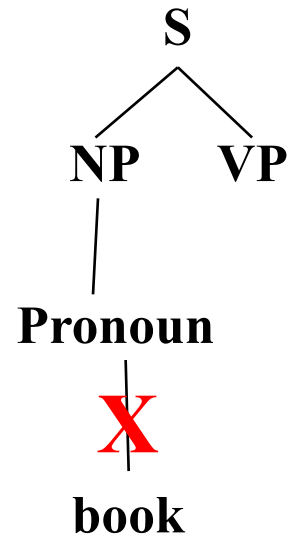
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Top Down Parsing

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$S \rightarrow \text{Aux } NP VP$

$S \rightarrow VP$

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$NP \rightarrow \text{Proper-Noun}$

$NP \rightarrow \text{Det Nominal}$

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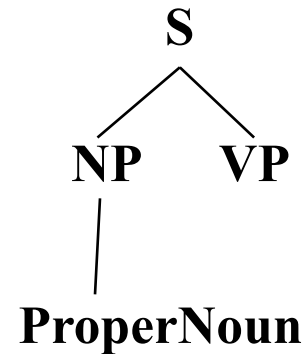
$\text{Nominal} \rightarrow \text{Nominal PP}$

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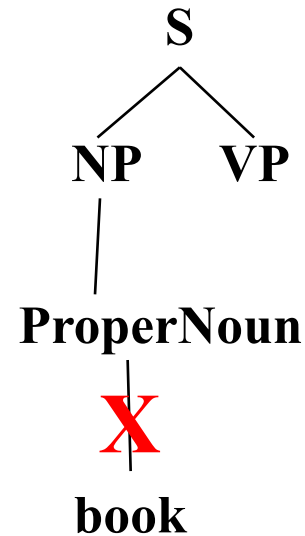
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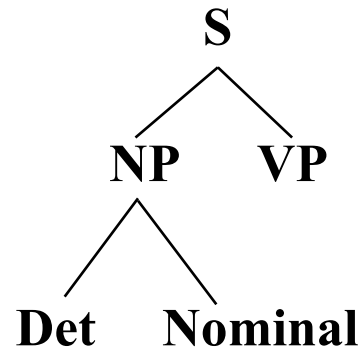
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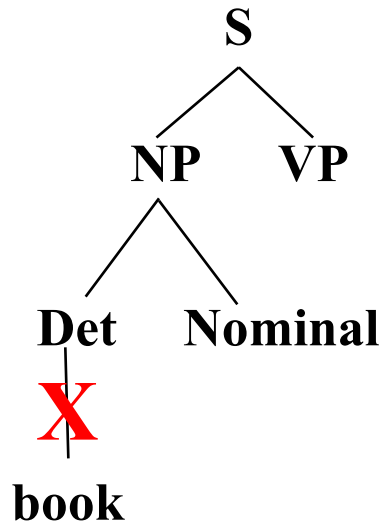
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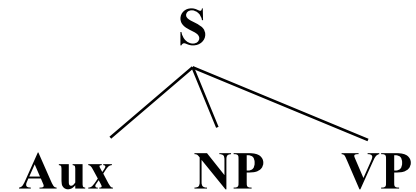
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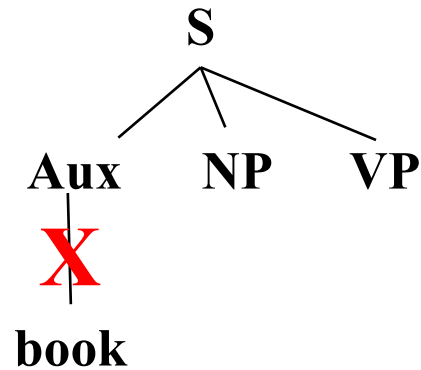
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S
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VP

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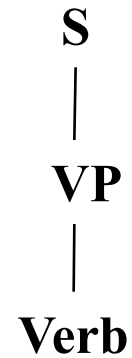
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S
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VP
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Verb
|
book

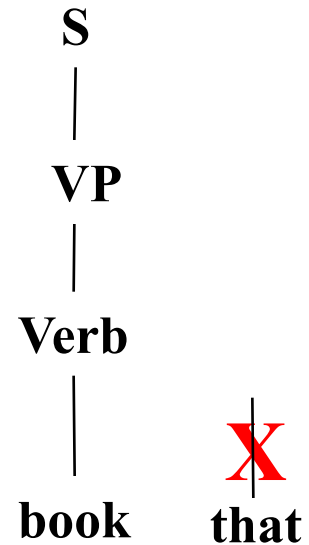
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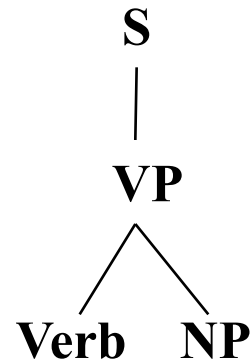
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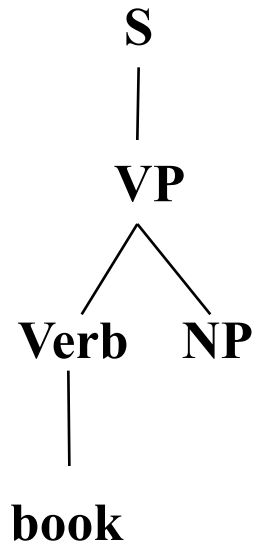
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Top Down Parsing

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$\text{Nominal} \rightarrow \text{Nominal Noun}$

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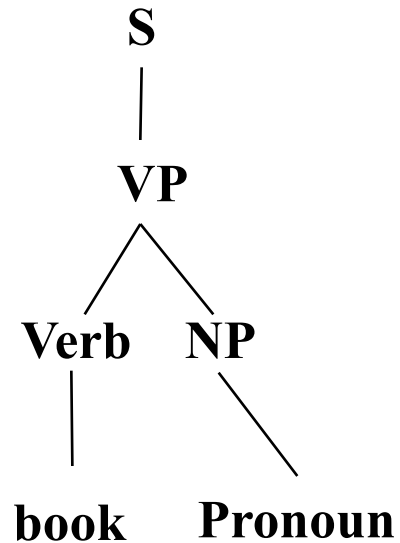
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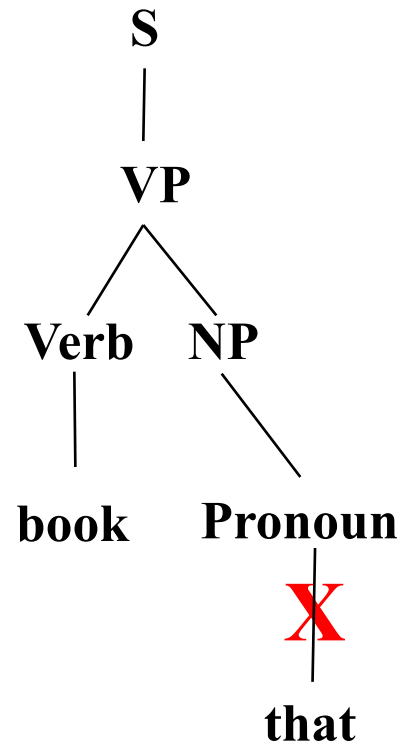
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$Aux \rightarrow does$

$Prep \rightarrow from \mid to \mid on \mid near \mid through$



Top Down Parsing

Grammar

$S \rightarrow NP VP$

$S \rightarrow Aux NP VP$

$S \rightarrow VP$

$NP \rightarrow Pronoun$

$NP \rightarrow Proper-Noun$

$NP \rightarrow Det Nominal$

$Nominal \rightarrow Noun$

$Nominal \rightarrow Nominal Noun$

$Nominal \rightarrow Nominal PP$

$VP \rightarrow Verb$

$VP \rightarrow Verb NP$

$VP \rightarrow VP PP$

$PP \rightarrow Prep NP$

Lexicon

$Det \rightarrow the \mid a \mid that \mid this$

$Noun \rightarrow book \mid flight \mid meal \mid money$

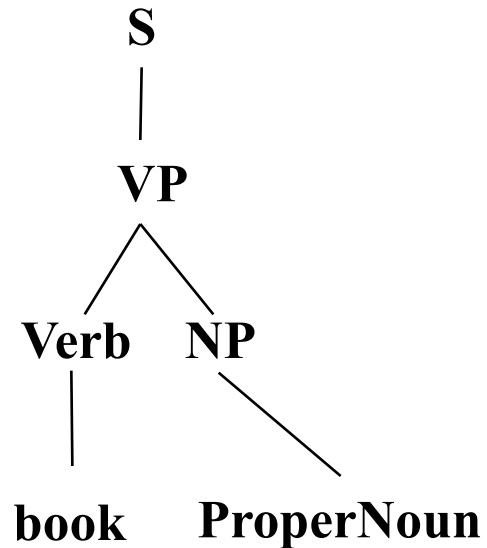
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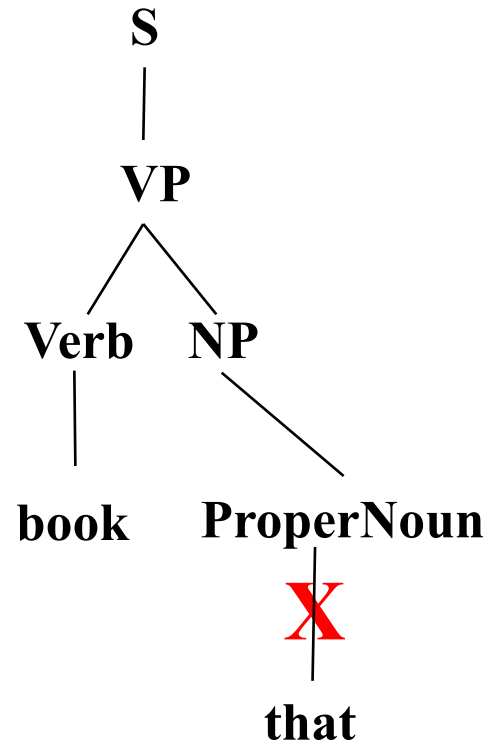
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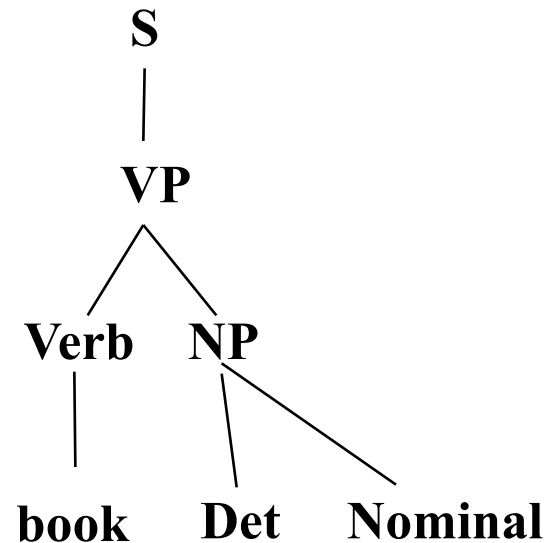
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Det $\rightarrow$ the | a | that | this

Noun $\rightarrow$ book | flight | meal | money

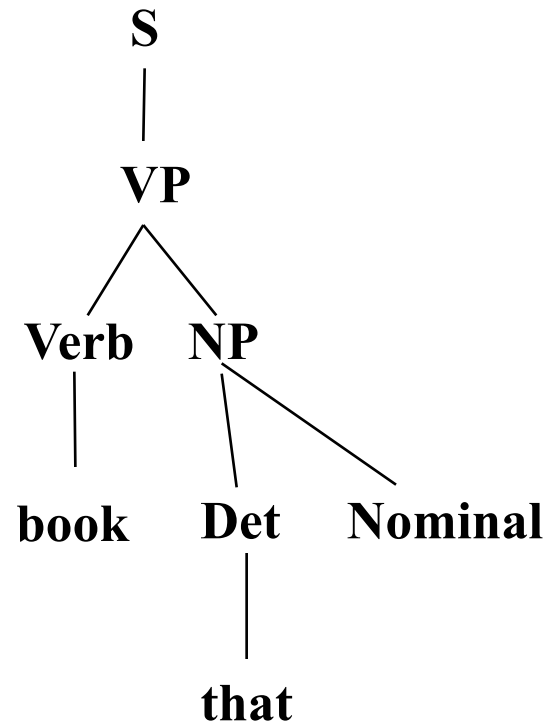
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Aux $\rightarrow$ does

Prep $\rightarrow$ from | to | on | near | through



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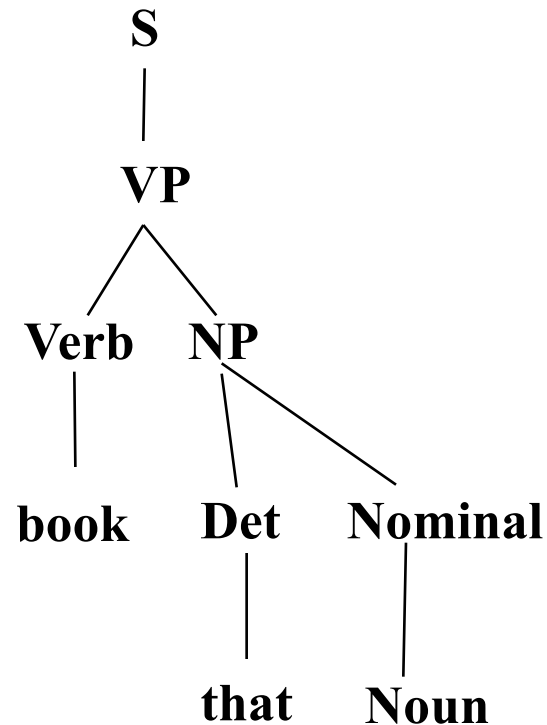
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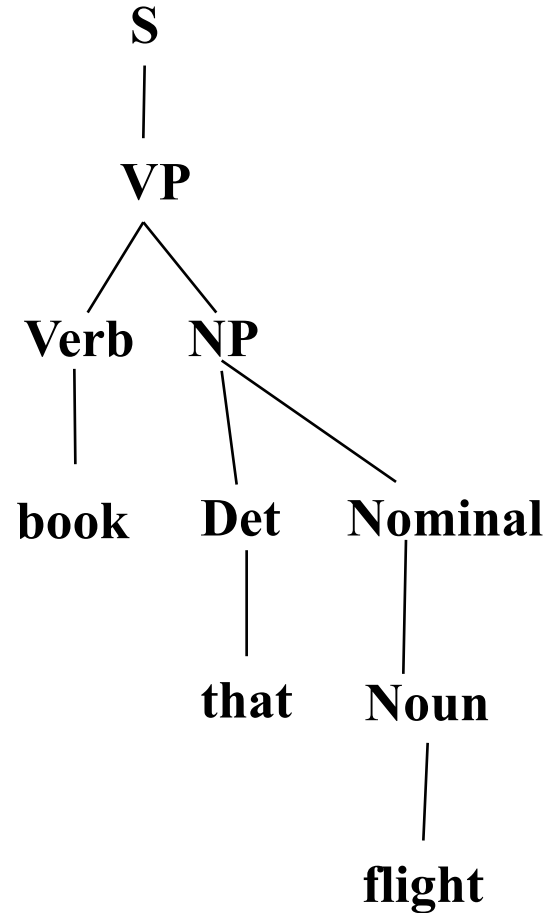
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book

that

flight

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Noun

book

that

flight

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Nominal

|

Noun

|

book

that

flight

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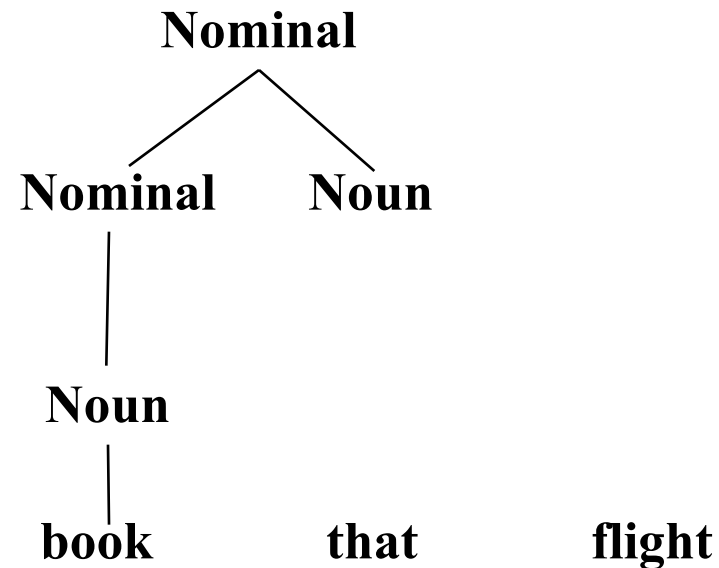
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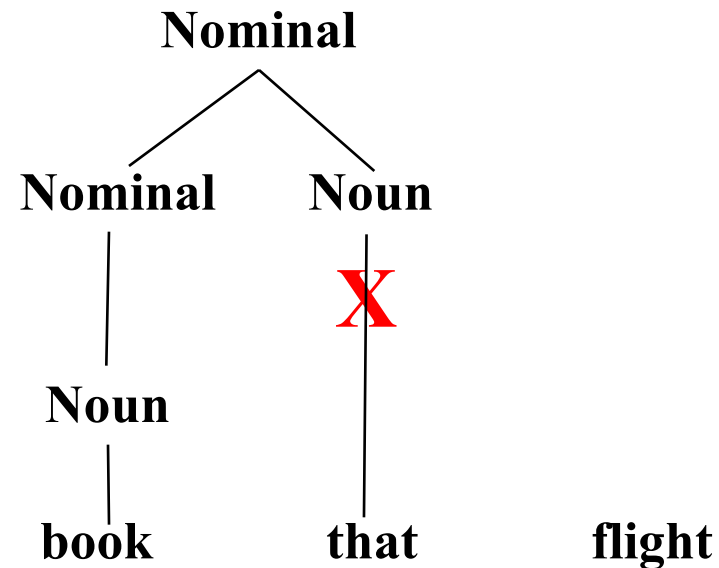
Bottom Up Parsing

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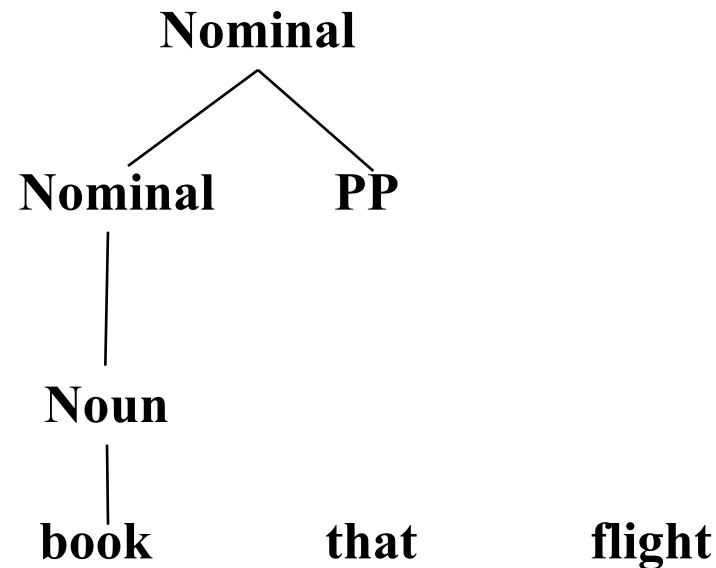
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$VP \rightarrow VP PP$

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Lexicon

Det $\rightarrow$ the | a | that | this

Noun $\rightarrow$ book | flight | meal | money

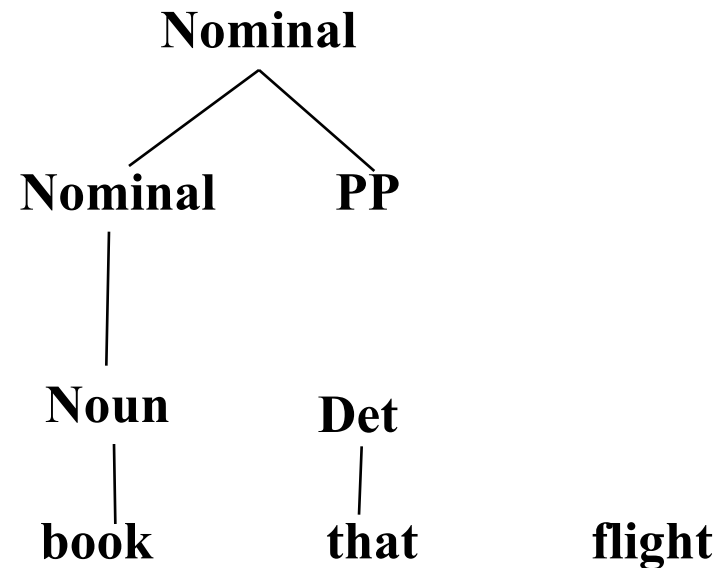
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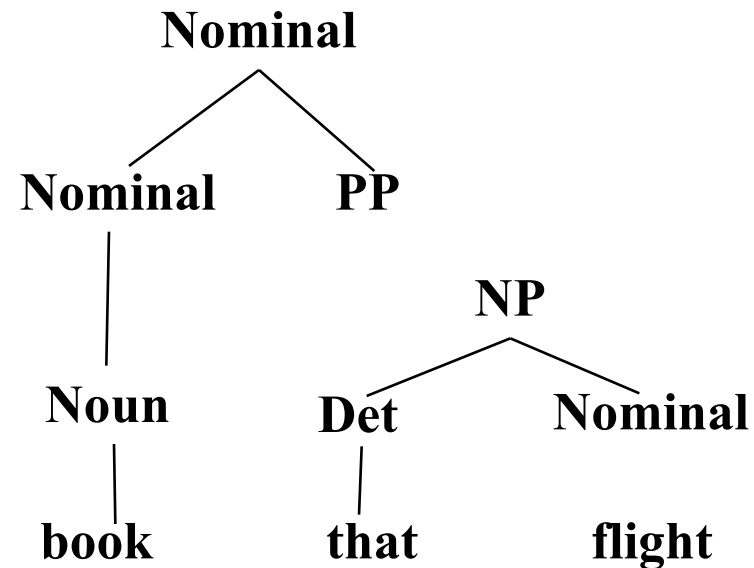
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Bottom Up Parsing

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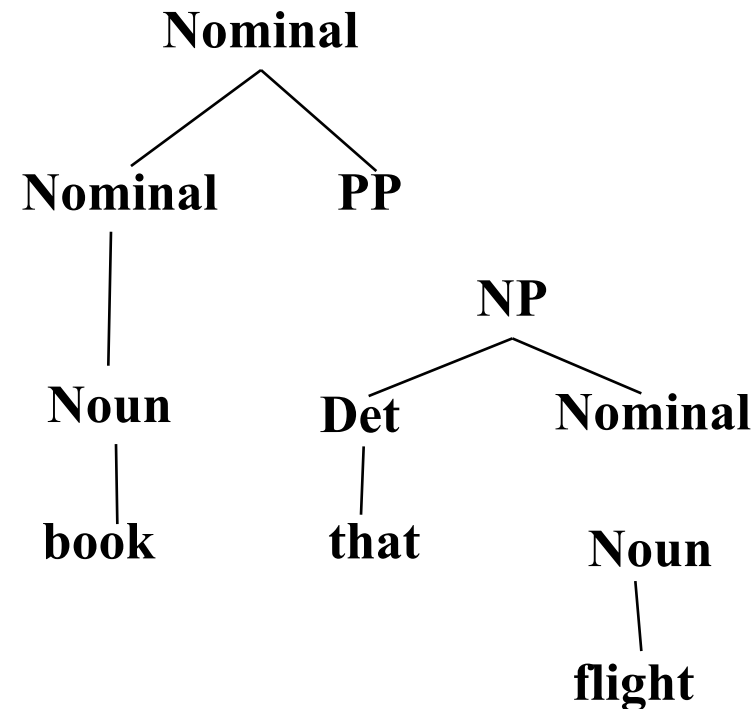
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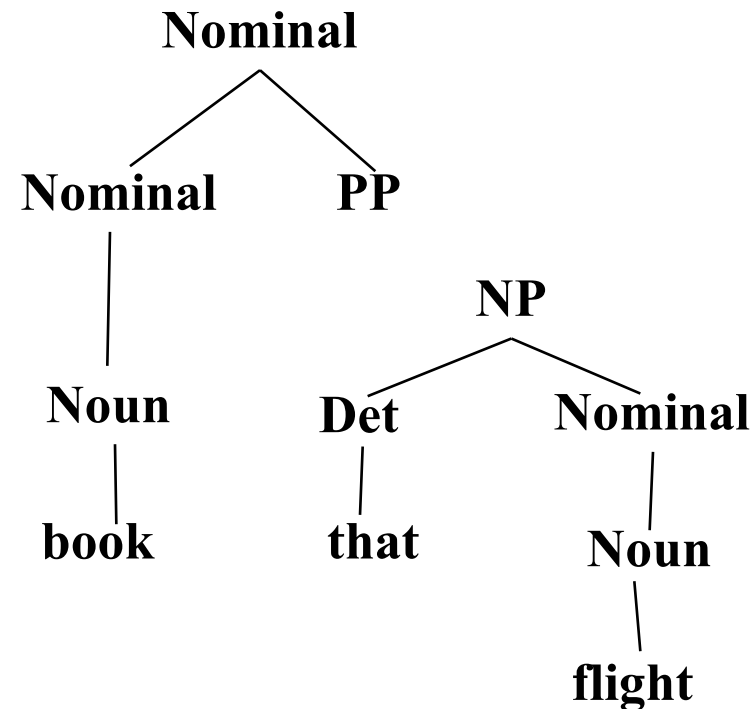
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Bottom Up Parsing

Grammar

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S → Aux NP VP

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NP → Det Nominal

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Nominal → Nominal PP

VP → Verb

VP → Verb NP

VP → VP PP

PP → Prep NP

Lexicon

Det → the | a | that | this

Noun → book | flight | meal | money

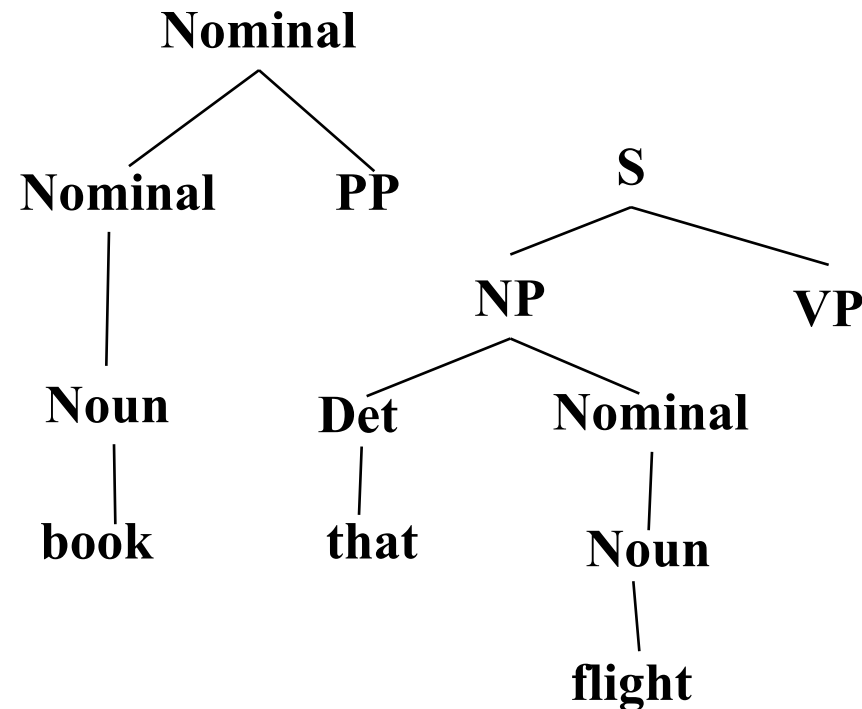
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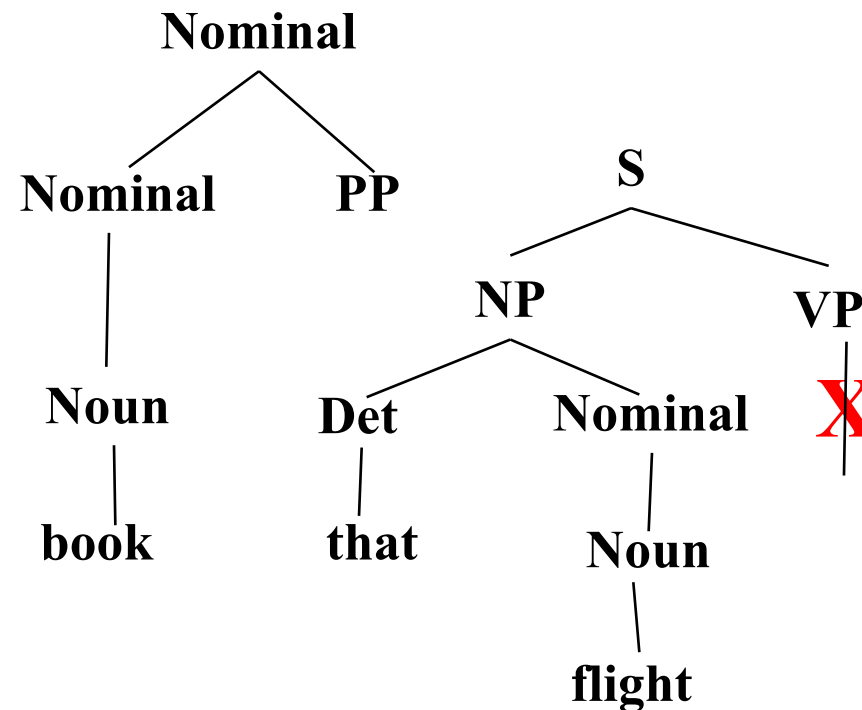
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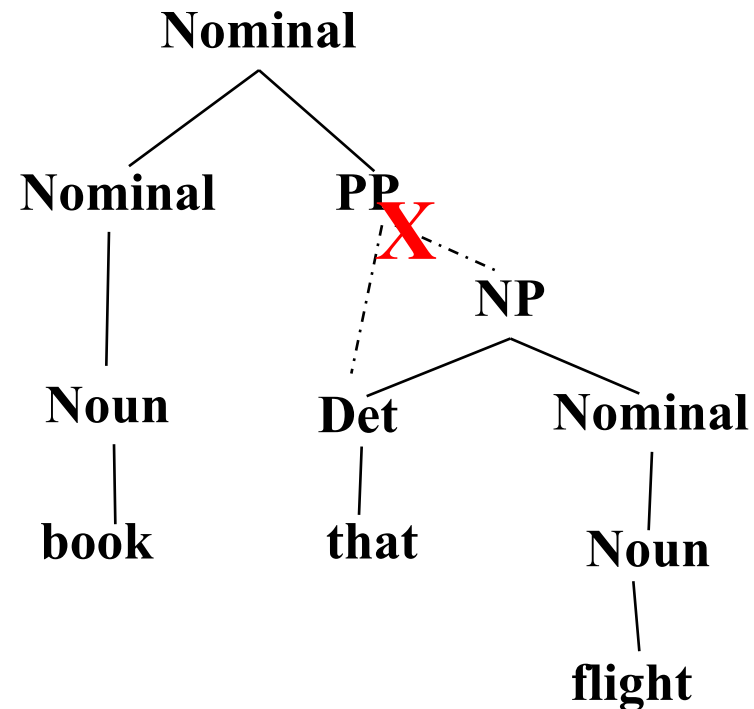
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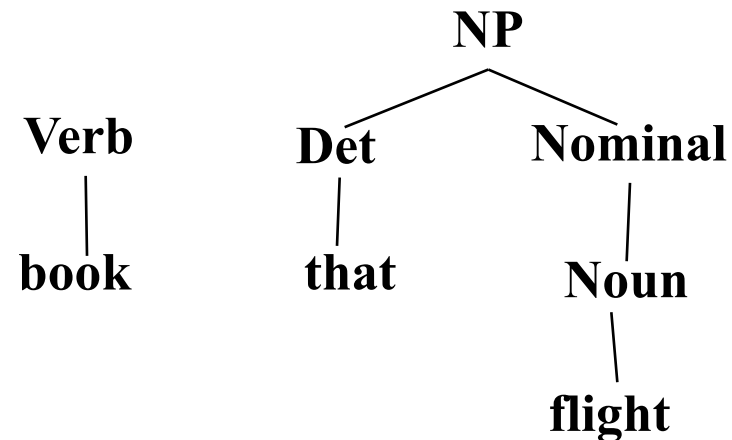
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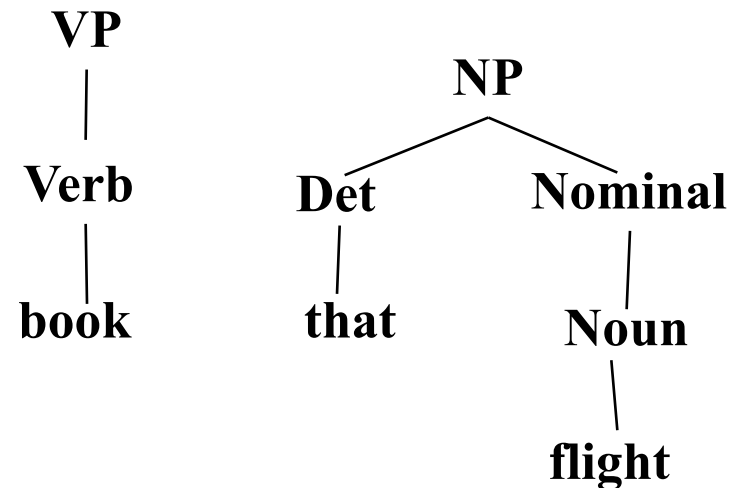
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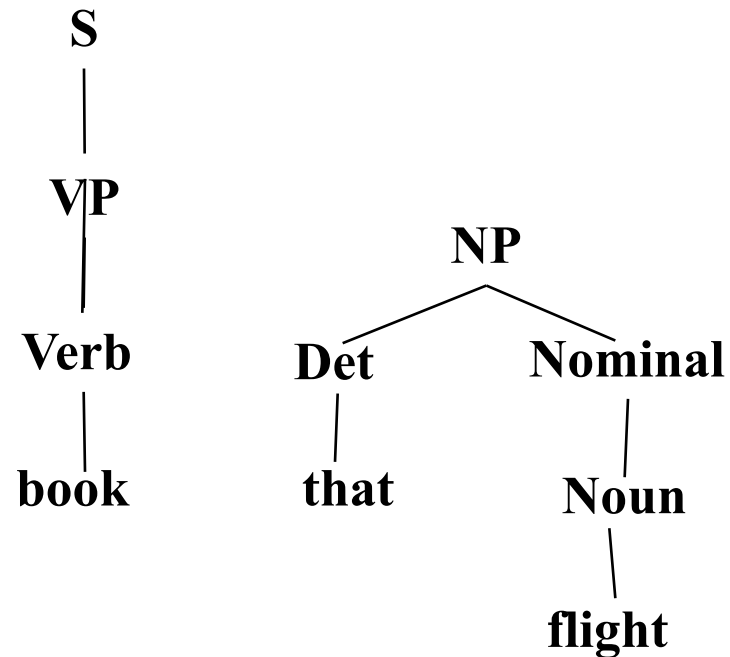
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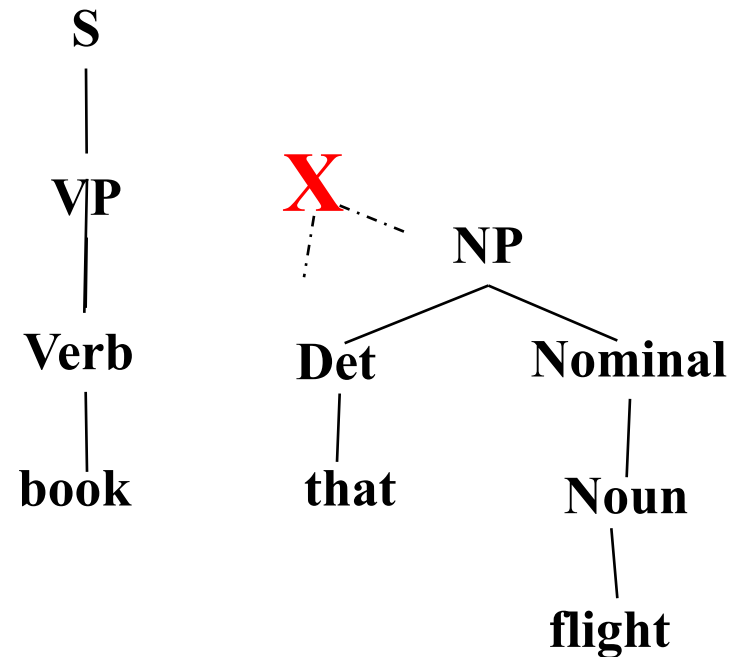
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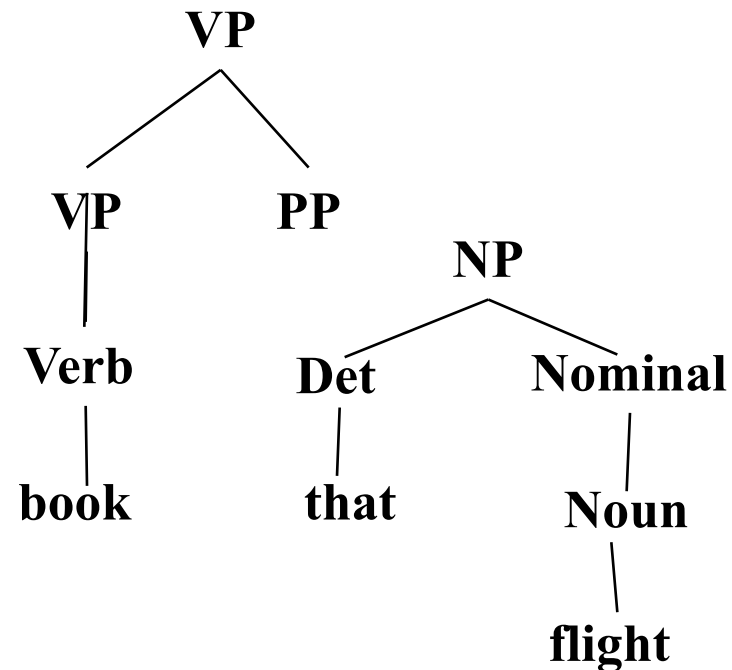
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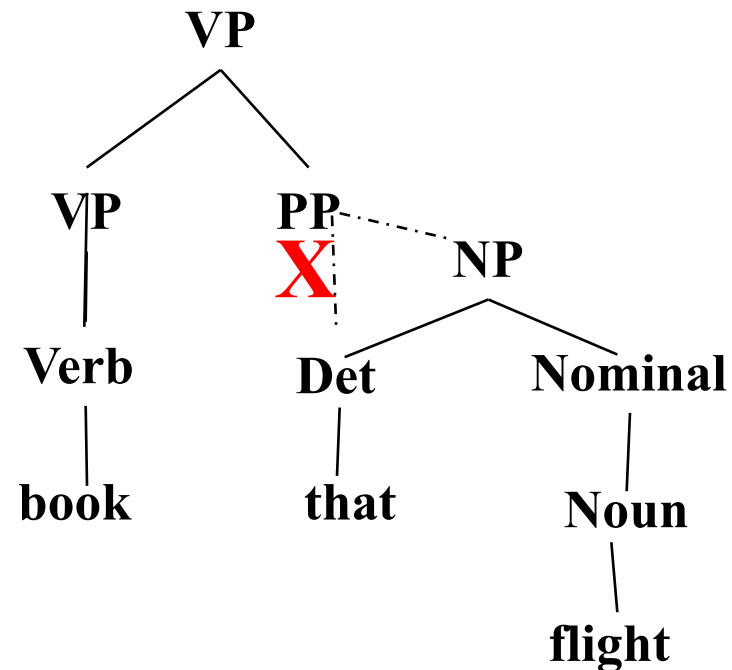
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$Pronoun \rightarrow I | he | she | me$

$Proper-Noun \rightarrow Houston | NWA$

$Aux \rightarrow does$

$Prep \rightarrow from | to | on | near | through$



Bottom Up Parsing

Grammar

$S \rightarrow NP VP$

$S \rightarrow Aux NP VP$

$S \rightarrow VP$

$NP \rightarrow Pronoun$

$NP \rightarrow Proper-Noun$

$NP \rightarrow Det Nominal$

$Nominal \rightarrow Noun$

$Nominal \rightarrow Nominal Noun$

$Nominal \rightarrow Nominal PP$

$VP \rightarrow Verb$

$VP \rightarrow Verb NP$

$VP \rightarrow VP PP$

$PP \rightarrow Prep NP$

Lexicon

$Det \rightarrow the \mid a \mid that \mid this$

$Noun \rightarrow book \mid flight \mid meal \mid money$

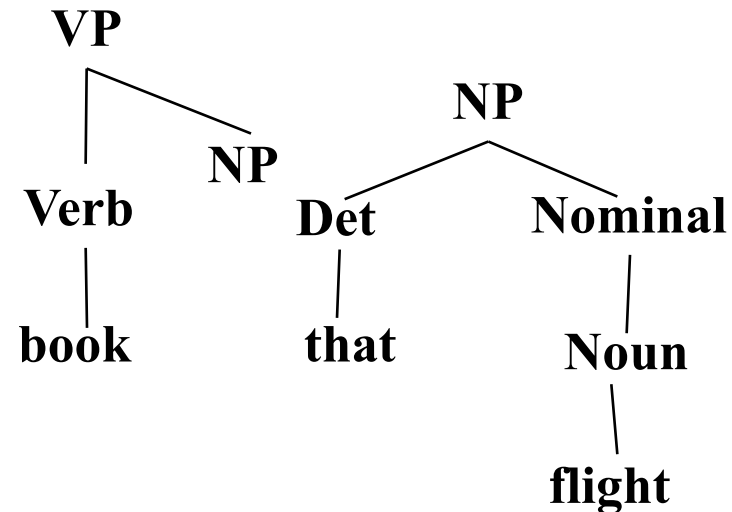
$Verb \rightarrow book \mid include \mid prefer$

$Pronoun \rightarrow I \mid he \mid she \mid me$

$Proper-Noun \rightarrow Houston \mid NWA$

$Aux \rightarrow does$

$Prep \rightarrow from \mid to \mid on \mid near \mid through$



Bottom Up Parsing

Grammar

$S \rightarrow NP VP$

$S \rightarrow Aux NP VP$

$S \rightarrow VP$

$NP \rightarrow Pronoun$

$NP \rightarrow Proper-Noun$

$NP \rightarrow Det Nominal$

$Nominal \rightarrow Noun$

$Nominal \rightarrow Nominal Noun$

$Nominal \rightarrow Nominal PP$

$VP \rightarrow Verb$

$VP \rightarrow Verb NP$

$VP \rightarrow VP PP$

$PP \rightarrow Prep NP$

Lexicon

$Det \rightarrow the \mid a \mid that \mid this$

$Noun \rightarrow book \mid flight \mid meal \mid money$

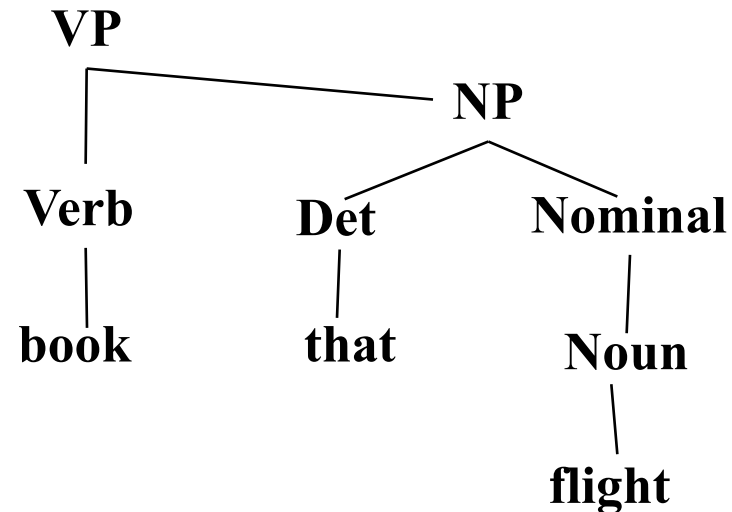
$Verb \rightarrow book \mid include \mid prefer$

$Pronoun \rightarrow I \mid he \mid she \mid me$

$Proper-Noun \rightarrow Houston \mid NWA$

$Aux \rightarrow does$

$Prep \rightarrow from \mid to \mid on \mid near \mid through$



Bottom Up Parsing

Grammar

$S \rightarrow NP VP$

$S \rightarrow Aux NP VP$

$S \rightarrow VP$

$NP \rightarrow Pronoun$

$NP \rightarrow Proper-Noun$

$NP \rightarrow Det Nominal$

$Nominal \rightarrow Noun$

$Nominal \rightarrow Nominal Noun$

$Nominal \rightarrow Nominal PP$

$VP \rightarrow Verb$

$VP \rightarrow Verb NP$

$VP \rightarrow VP PP$

$PP \rightarrow Prep NP$

Lexicon

$Det \rightarrow the \mid a \mid that \mid this$

$Noun \rightarrow book \mid flight \mid meal \mid money$

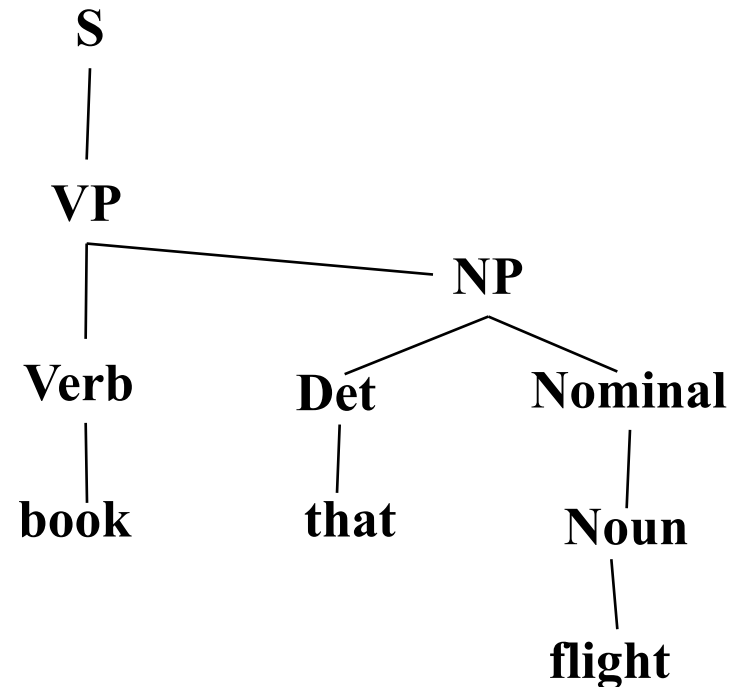
$Verb \rightarrow book \mid include \mid prefer$

$Pronoun \rightarrow I \mid he \mid she \mid me$

$Proper-Noun \rightarrow Houston \mid NWA$

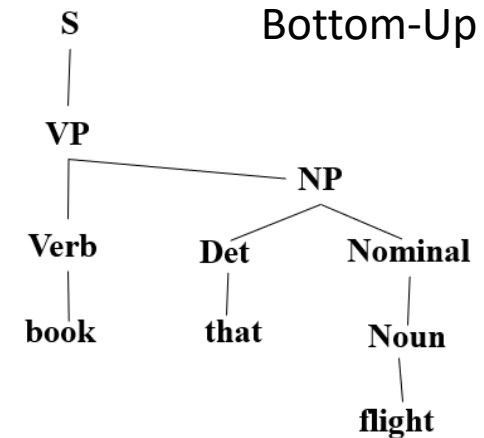
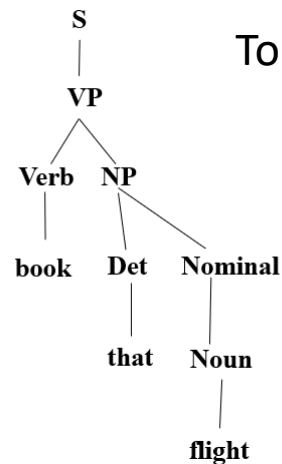
$Aux \rightarrow does$

$Prep \rightarrow from \mid to \mid on \mid near \mid through$



Top Down vs. Bottom Up

- Top down **never** explores options that will not lead to a full parse, but can **explore many options that never connect to the actual sentence.**
- Bottom up **never** explores options that do not connect to the actual sentence but **can explore options that can never lead to a full parse.**
- Relative amounts of wasted search depend on how much the grammar branches in each direction.



Overview

1. Syntax
2. Syntactic Parsing
3. **CKY Parsing**
4. Statistical Parsing
5. Probabilistic CKY Parsing
6. PCFG Training
7. Lexicalized PCFGs
8. Evaluating model performance
9. Alternatives
10. New Developments

Dynamic Programming Parsing

- To avoid extensive repeated work, must cache intermediate results, i.e. completed phrases.
- Caching (memorizing) critical to obtaining a polynomial time parsing (recognition) algorithm for CFGs.
- Dynamic programming algorithms based on both top-down and bottom-up search can achieve $O(n^3)$ recognition time where n is the length of the input string.

Dynamic Programming Parsing Methods

- **CKY** (Cocke-Kasami-Younger) algorithm based on bottom-up parsing and requires first normalizing the grammar.
- **Earley parser** is based on top-down parsing and does not require normalizing grammar but is more complex.
- More generally, **chart parsers** retain completed phrases in a chart and can combine top-down and bottom-up search.

CKY chart parsing algorithm

- Bottom-up parsing:
 - Start with the words
- Dynamic programming
 - Save the results in a table/chart
 - Re-use these results in finding larger constituents
- First grammar must be converted to **Chomsky normal form (CNF)** in which productions must have either:
 - Exactly **2 non-terminal** symbols on the RHS
 - Or **1 terminal symbol** (lexicon rules).

Chomsky Normal Form

- The right-hand side of a standard CFG rules can have an arbitrary number of symbols (terminals and non-terminals)

VP $\rightarrow$ ADV eat NP



- A CFG in Chomsky Normal Form (CNF) allows only two kinds of right-hand sides:
 - Two non-terminals: VP $\rightarrow$ ADV VP
 - One terminal: VP $\rightarrow$ eat
- Any CFG can be transformed into an equivalent CFG in CNF by introducing new, rule-specific dummy non-terminals (VP₁, VP₂, ...)

VP $\rightarrow$ ADV VP₁

VP₁ $\rightarrow$ ADV VP₂

VP₂ $\rightarrow$ eat



Revisit - Our toy CFG to convert to CNF

Grammar

S → **NP VP**
S → **Aux NP VP**
S → **VP**
NP → **Pronoun**
NP → **Proper-Noun**
NP → **Det Nominal**
Nominal → **Noun**
Nominal → **Nominal Noun**
Nominal → **Nominal PP**
VP → **Verb**
VP → **Verb NP**
VP → **VP PP**
PP → **Prep NP**

Lexicon

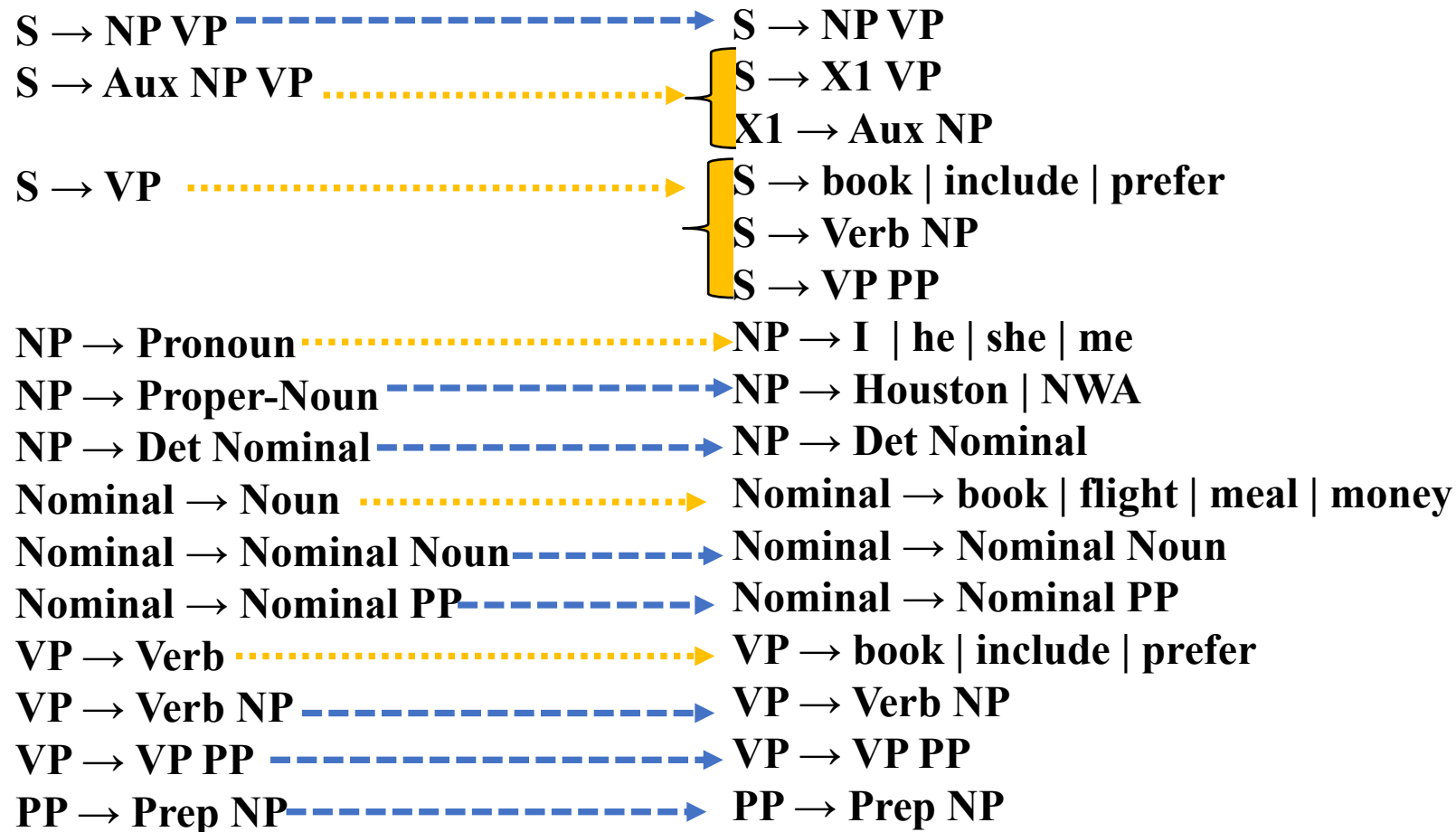
Det → **the | a | that | this**
Noun → **book | flight | meal | money**
Verb → **book | include | prefer**
Pronoun → **I | he | she | me**
Proper-Noun → **Houston | NWA**
Aux → **does**
Prep → **from | to | on | near | through**

Lexicon rules don't need conversion,
they are already in CNF format.

Chomsky Normal Form (CNF) Conversion

Original Grammar

Chomsky Normal Form



CKY Parser

table[i,j] contains all constituents (non-terminals) words $i + 1$ through j

Find all possible split point k , such that

table[i,j] $\rightarrow$ table[i,k] table[k,j]

function CKY-PARSE(words, grammar) **returns** table

for $j \leftarrow$ from 1 to LENGTH(words) **do**

for all $\{A \mid A \rightarrow \text{words}[j] \in \text{grammar}\}$

$\text{table}[j - 1, j] \leftarrow \text{table}[j - 1, j] \cup A$

for $i \leftarrow$ from $j - 2$ down to 0 **do**

for $k \leftarrow i + 1$ to $j - 1$ **do**

for all $\{A \mid A \rightarrow BC \in \text{grammar} \text{ and } B \in \text{table}[i, k] \text{ and } C \in \text{table}[k, j]\}$

$\text{table}[i, j] \leftarrow \text{table}[i, j] \cup A$

Populate the
terminal (word)

Partition into two parts at
every possible position k

	Book j=1	the 2	flight 3	through 4	Houston 5
i=0	book	took the			
1		the	the flight		
2			flight	flight through	
3				through	through Houston
4					Houston

The root (S)

Book the flight
through Houston

table[1,3] $\rightarrow$ table[1,2] table[2,1]

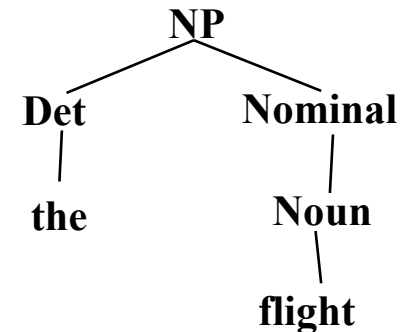


Figure 13.5 The CKY algorithm.

CKY Parser

Length = 1

	Book	the	flight	through	Houston
	j= 1	2	3	4	5
i= 0	S, VP, Verb, Nominal, Noun				
1		Det			
2			Nominal, Noun		
3				Det	
4					NP, Proper- Noun

R1: $S \rightarrow NP VP$

R2: $S \rightarrow X1 VP$

R3: $X1 \rightarrow Aux NP$

$S \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R4: $S \rightarrow Verb NP$

R5: $S \rightarrow VP PP$

$NP \rightarrow I \mid he \mid she \mid me$

$NP \rightarrow \text{Houston} \mid \text{NWA}$

R6: $NP \rightarrow Det Nominal$

$Nominal \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

R7: $Nominal \rightarrow Nominal Noun$

R8: $Nominal \rightarrow Nominal PP$

$VP \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R9: $VP \rightarrow Verb NP$

R10: $VP \rightarrow VP PP$

R11: $PP \rightarrow Prep NP$

$Det \rightarrow \text{the} \mid a \mid that \mid this$

$Noun \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

$Verb \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

$Pronoun \rightarrow I \mid he \mid she \mid me$

$Proper-Noun \rightarrow \text{Houston} \mid \text{NWA}$

$Aux \rightarrow \text{does}$

$Prep \rightarrow \text{from} \mid \text{to} \mid \text{on} \mid \text{near} \mid \text{through}$

CKY Parser

Length = 2

	Book	the	flight	through	Houston
	j= 1	2	3	4	5
i= 0	S, VP, Verb, Nominal, Noun	None			
1		Det	NP (R6, 1)		
2			Nominal, Noun	None	
3				Det	PP (R11, 4)
4					NP, Proper-Noun

R1: $S \rightarrow NP VP$

R2: $S \rightarrow X1 VP$

R3: $X1 \rightarrow Aux NP$

$S \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R4: $S \rightarrow Verb NP$

R5: $S \rightarrow VP PP$

$NP \rightarrow I \mid he \mid she \mid me$

$NP \rightarrow \text{Houston} \mid \text{NWA}$

R6: $NP \rightarrow Det Nominal$

$Nominal \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

R7: $Nominal \rightarrow Nominal Noun$

R8: $Nominal \rightarrow Nominal PP$

$VP \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R9: $VP \rightarrow Verb NP$

R10: $VP \rightarrow VP PP$

R11: $PP \rightarrow Prep NP$

$Det \rightarrow \text{the} \mid a \mid that \mid this$

$Noun \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

$Verb \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

$Pronoun \rightarrow I \mid he \mid she \mid me$

$Proper-Noun \rightarrow \text{Houston} \mid \text{NWA}$

$Aux \rightarrow \text{does}$

$Prep \rightarrow \text{from} \mid \text{to} \mid \text{on} \mid \text{near} \mid \text{through}$

CKY Parser

Length = 3

	Book	the	flight	through	Houston
	j= 1	2	3	4	5
i= 0	S, VP, Verb, Nominal, Noun	None	VP (R9, 1)		
1		Det	NP (R6, 1)		
2			Nominal, Noun	None	
3				Det	PP (R11, 4)
4					NP, Proper- Noun

R1: $S \rightarrow NP VP$

R2: $S \rightarrow X1 VP$

R3: $X1 \rightarrow Aux NP$

$S \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R4: $S \rightarrow Verb NP$

R5: $S \rightarrow VP PP$

$NP \rightarrow I \mid he \mid she \mid me$

$NP \rightarrow \text{Houston} \mid \text{NWA}$

R6: $NP \rightarrow Det Nominal$

$Nominal \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

R7: $Nominal \rightarrow Nominal Noun$

R8: $Nominal \rightarrow Nominal PP$

$VP \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R9: $VP \rightarrow Verb NP$

R10: $VP \rightarrow VP PP$

R11: $PP \rightarrow Prep NP$

$Det \rightarrow \text{the} \mid a \mid that \mid this$

$Noun \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

$Verb \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

$Pronoun \rightarrow I \mid he \mid she \mid me$

$Proper-Noun \rightarrow \text{Houston} \mid \text{NWA}$

$Aux \rightarrow \text{does}$

$Prep \rightarrow \text{from} \mid \text{to} \mid \text{on} \mid \text{near} \mid \text{through}$

CKY Parser

Length = 3

	Book	the	flight	through	Houston
	j= 1	2	3	4	5
i= 0	S, VP, Verb, Nominal, Noun	None	VP (R9, 1) S (R4, 1)		
1		Det	NP (R6, 1)	None	
2			Nominal, Noun	None	Nominal (R8, 3)
3				Det	PP (R11, 4)
4					NP, Proper- Noun

R1: $S \rightarrow NP VP$

R2: $S \rightarrow X1 VP$

R3: $X1 \rightarrow Aux NP$

$S \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R4: $S \rightarrow Verb NP$

R5: $S \rightarrow VP PP$

$NP \rightarrow I \mid he \mid she \mid me$

$NP \rightarrow \text{Houston} \mid \text{NWA}$

R6: $NP \rightarrow Det Nominal$

$Nominal \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

R7: $Nominal \rightarrow Nominal Noun$

R8: $Nominal \rightarrow Nominal PP$

$VP \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R9: $VP \rightarrow Verb NP$

R10: $VP \rightarrow VP PP$

R11: $PP \rightarrow Prep NP$

$Det \rightarrow \text{the} \mid a \mid that \mid this$

$Noun \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

$Verb \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

$Pronoun \rightarrow I \mid he \mid she \mid me$

$Proper-Noun \rightarrow \text{Houston} \mid \text{NWA}$

$Aux \rightarrow \text{does}$

$Prep \rightarrow \text{from} \mid \text{to} \mid \text{on} \mid \text{near} \mid \text{through}$

CKY Parser

Length = 4

	Book	the	flight	through	Houston
	j= 1	2	3	4	5
i= 0	S, VP, Verb, Nominal, Noun	None	VP (R9, 1) S (R4, 1)	None	
1		Det	NP (R6, 1)	None	NP (R6, 2)
2			Nominal, Noun	None	Nominal (R8, 3)
3				Det	PP (R11, 4)
4					NP, Proper- Noun

R1: $S \rightarrow NP VP$

R2: $S \rightarrow X1 VP$

R3: $X1 \rightarrow Aux NP$

$S \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R4: $S \rightarrow Verb NP$

R5: $S \rightarrow VP PP$

$NP \rightarrow I \mid he \mid she \mid me$

$NP \rightarrow \text{Houston} \mid \text{NWA}$

R6: $NP \rightarrow Det Nominal$

$Nominal \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

R7: $Nominal \rightarrow Nominal Noun$

R8: $Nominal \rightarrow Nominal PP$

$VP \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R9: $VP \rightarrow Verb NP$

R10: $VP \rightarrow VP PP$

R11: $PP \rightarrow Prep NP$

$Det \rightarrow \text{the} \mid a \mid that \mid this$

$Noun \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

$Verb \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

$Pronoun \rightarrow I \mid he \mid she \mid me$

$Proper-Noun \rightarrow \text{Houston} \mid \text{NWA}$

$Aux \rightarrow \text{does}$

$Prep \rightarrow \text{from} \mid \text{to} \mid \text{on} \mid \text{near} \mid \text{through}$

CKY Parser

Length = 5

	Book	the	flight	through	Houston
	j= 1	2	3	4	5
i= 0	S, VP, Verb, Nominal, Noun	None	VP (R9, 1) S (R4, 1)	None	S (R5, 3) VP (R10, 3) S (R4, 1) VP (R9, 1)
1		Det	NP (R6, 1)	None	NP (R6, 2)
2			Nominal, Noun	None	Nominal (R8, 3)
3				Det	PP (R11, 4)
4					NP, Proper- Noun

R1: $S \rightarrow NP VP$

R2: $S \rightarrow X1 VP$

R3: $X1 \rightarrow Aux NP$

$S \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R4: $S \rightarrow Verb NP$

R5: $S \rightarrow VP PP$

$NP \rightarrow I \mid he \mid she \mid me$

$NP \rightarrow \text{Houston} \mid \text{NWA}$

R6: $NP \rightarrow Det Nominal$

$Nominal \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

R7: $Nominal \rightarrow Nominal Noun$

R8: $Nominal \rightarrow Nominal PP$

$VP \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R9: $VP \rightarrow Verb NP$

R10: $VP \rightarrow VP PP$

R11: $PP \rightarrow Prep NP$

$Det \rightarrow \text{the} \mid a \mid that \mid this$

$Noun \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

$Verb \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

$Pronoun \rightarrow I \mid he \mid she \mid me$

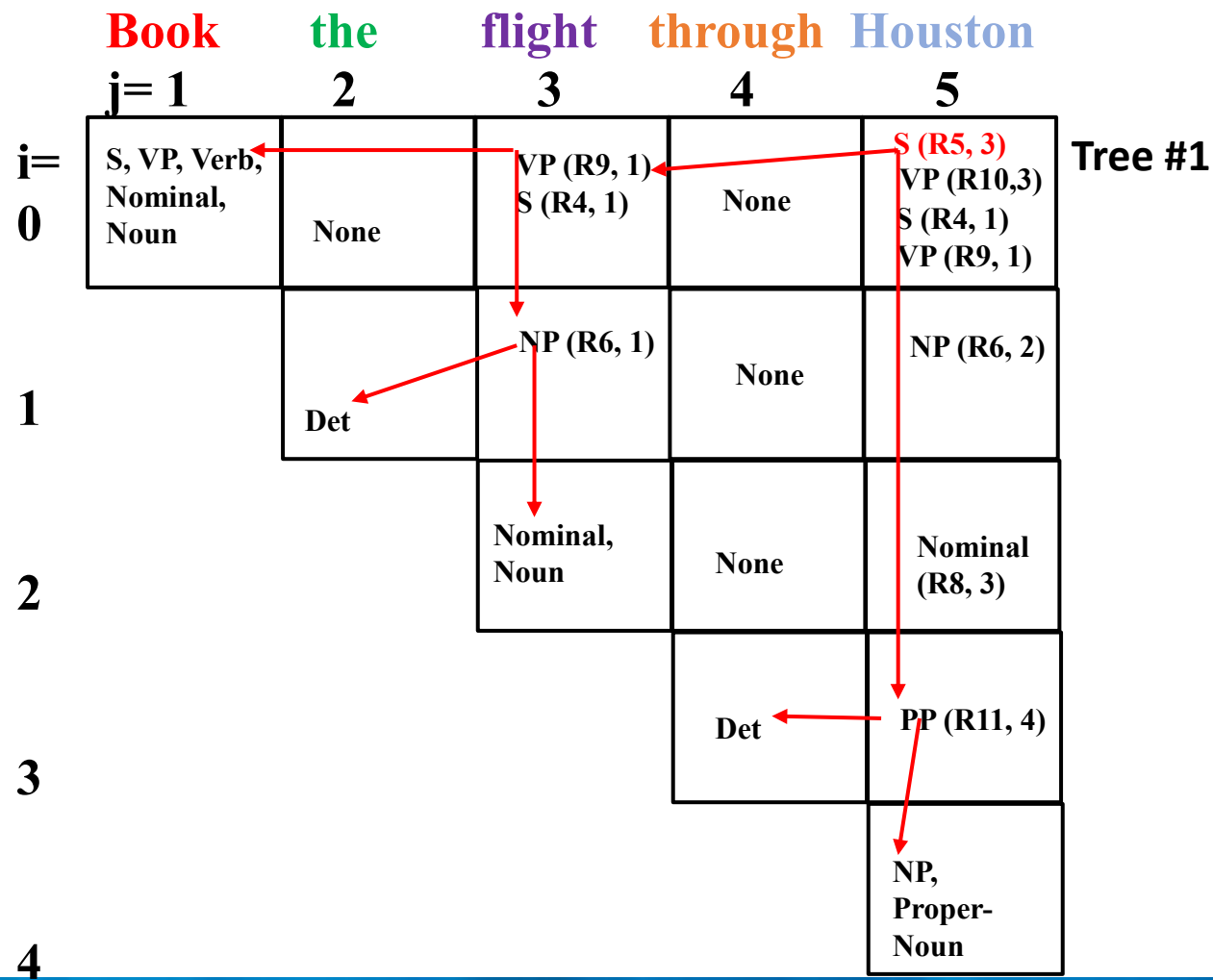
$Proper-Noun \rightarrow \text{Houston} \mid \text{NWA}$

$Aux \rightarrow \text{does}$

$Prep \rightarrow \text{from} \mid \text{to} \mid \text{on} \mid \text{near} \mid \text{through}$

CKY Parser

Length = 5



R1: $S \rightarrow NP VP$

R2: $S \rightarrow X1 VP$

R3: $X1 \rightarrow Aux NP$

$S \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R4: $S \rightarrow Verb NP$

R5: $S \rightarrow VP PP$

$NP \rightarrow I \mid he \mid she \mid me$

$NP \rightarrow \text{Houston} \mid \text{NWA}$

R6: $NP \rightarrow Det Nominal$

$Nominal \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

R7: $Nominal \rightarrow Nominal Noun$

R8: $Nominal \rightarrow Nominal PP$

$VP \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

R9: $VP \rightarrow Verb NP$

R10: $VP \rightarrow VP PP$

R11: $PP \rightarrow Prep NP$

$Det \rightarrow \text{the} \mid a \mid \text{that} \mid \text{this}$

$Noun \rightarrow \text{book} \mid \text{flight} \mid \text{meal} \mid \text{money}$

$Verb \rightarrow \text{book} \mid \text{include} \mid \text{prefer}$

$Pronoun \rightarrow I \mid he \mid she \mid me$

$Proper-Noun \rightarrow \text{Houston} \mid \text{NWA}$

$Aux \rightarrow \text{does}$

$Prep \rightarrow \text{from} \mid \text{to} \mid \text{on} \mid \text{near} \mid \text{through}$

CKY Parser

Length = 5

	Book	the	flight	through	Houston
	j= 1	2	3	4	5
i= 0	S, VP, Verb, Nominal, Noun	None	VP (R9, 1) S (R4, 1)	None	S (R5, 3) VP (R10,3) S (R4, 1) VP (R9, 1)
1		Det	NP (R6, 1)	None	NP (R6, 2)
2			Nominal, Noun	None	Nominal (R8, 3)
3				Det	PP (R11, 4)
4					NP, Proper- Noun

Tree #2

- R1: S → NP VP
 - R2: S → X1 VP
 - R3: X1 → Aux NP
S → **book** | include | prefer
 - R4: S → Verb NP**
 - R5: S → VP PP
NP → I | he | she | me
NP → **Houston** | NWA
 - R6: NP → Det Nominal**
Nominal → **book** | **flight** | meal | money
 - R7: Nominal → Nominal Noun
 - R8: Nominal → Nominal PP**
VP → **book** | include | prefer
 - R9: VP → Verb NP
 - R10: VP → VP PP
 - R11: PP → Prep NP**
- Det → **the** | a | that | this
Noun → **book** | **flight** | meal | money
Verb → **book** | include | prefer
Pronoun → I | he | she | me
Proper-Noun → **Houston** | NWA
Aux → does
Prep → from | to | on | near | **through**

Complexity of CKY (recognition)

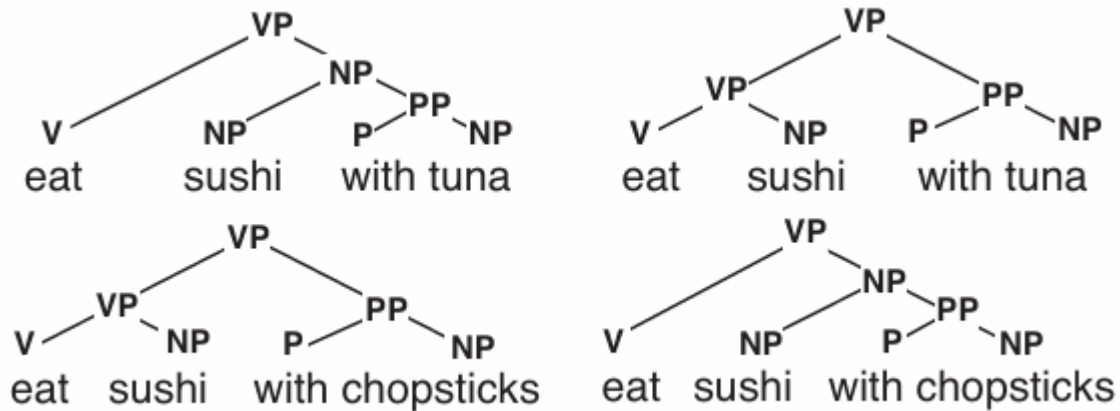
- There are $(n(n+1)/2) = O(n^2)$ cells
- Filling each cell requires looking at every possible split point between the two non-terminals needed to introduce a new phrase.
- There are $O(n)$ possible split points.
- Total time complexity is $O(n^3)$

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Grammars are ambiguous

- A grammar might generate multiple trees for a sentence



- What's the most likely parse tree τ for sentence S ?
- We need a model of $P(\tau|S)$

Statistical Parsing

- Statistical parsing uses a probabilistic model of syntax in order to assign probabilities to each parse tree.
- Provides principled approach to resolving syntactic ambiguity.
- Allows supervised learning of parsers from tree-banks of parse trees provided by human linguists.

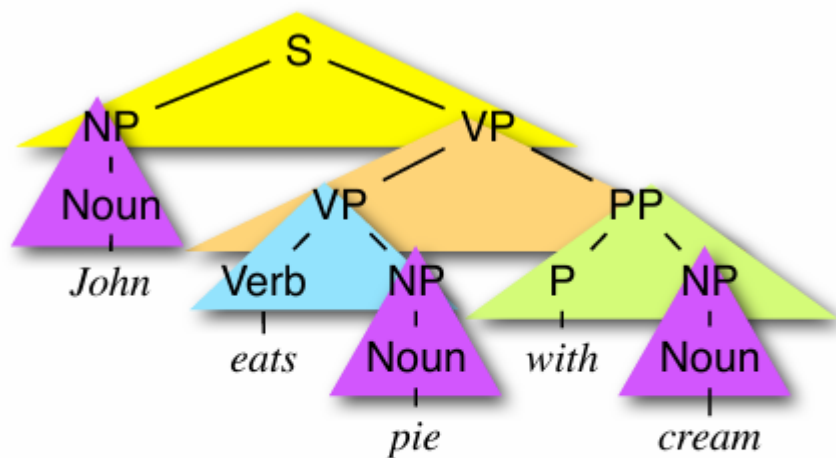
Probabilistic Context Free Grammar (PCFG)

- A PCFG is a probabilistic version of a CFG where each production has a probability.
- For every nonterminal X , define a probability distribution $P(X \rightarrow \alpha \mid X)$ over all rules with the same LHS symbol X .
- Probabilities of all productions rewriting a given non-terminal must add to 1, defining a distribution for each non-terminal.

S	$\rightarrow$	NP VP	0.8
S	$\rightarrow$	S conj S	0.2
NP	$\rightarrow$	Noun	0.2
NP	$\rightarrow$	Det Noun	0.4
NP	$\rightarrow$	NP PP	0.2
NP	$\rightarrow$	NP conj NP	0.2
VP	$\rightarrow$	Verb	0.4
VP	$\rightarrow$	Verb NP	0.3
VP	$\rightarrow$	Verb NP NP	0.1
VP	$\rightarrow$	VP PP	0.2
PP	$\rightarrow$	P NP	1.0

Computing $P(\tau)$ with a PCFG

- Assume productions for each node are chosen independently.
- The probability of a tree τ is the product of the probabilities of all its rules



$$P(\tau) = 0.8 \times 0.3 \times 0.2 \times 1.0 \times 0.2^3 = 0.00384$$

S	→ NP VP	0.8
S	→ S conj S	0.2
NP	→ Noun	0.2
NP	→ Det Noun	0.4
NP	→ NP PP	0.2
NP	→ NP conj NP	0.2
VP	→ Verb	0.4
VP	→ Verb NP	0.3
VP	→ Verb NP NP	0.1
VP	→ VP PP	0.2
PP	→ P NP	1.0

Revisit Our Toy PCFG

Grammar	Prob
$S \rightarrow NP VP$	0.8
$S \rightarrow Aux NP VP$	0.1
$S \rightarrow VP$	0.1
$NP \rightarrow Pronoun$	0.2
$NP \rightarrow Proper-Noun$	0.2
$NP \rightarrow Det Nominal$	0.6
$Nominal \rightarrow Noun$	0.3
$Nominal \rightarrow Nominal Noun$	0.2
$Nominal \rightarrow Nominal PP$	0.5
$VP \rightarrow Verb$	0.2
$VP \rightarrow Verb NP$	0.5
$VP \rightarrow VP PP$	0.3
$PP \rightarrow Prep NP$	1.0

0.8
 0.1
 0.1
 0.2
 0.2
 0.6
 0.3
 0.2
 0.5
 0.2
 0.5
 0.3
 1.0

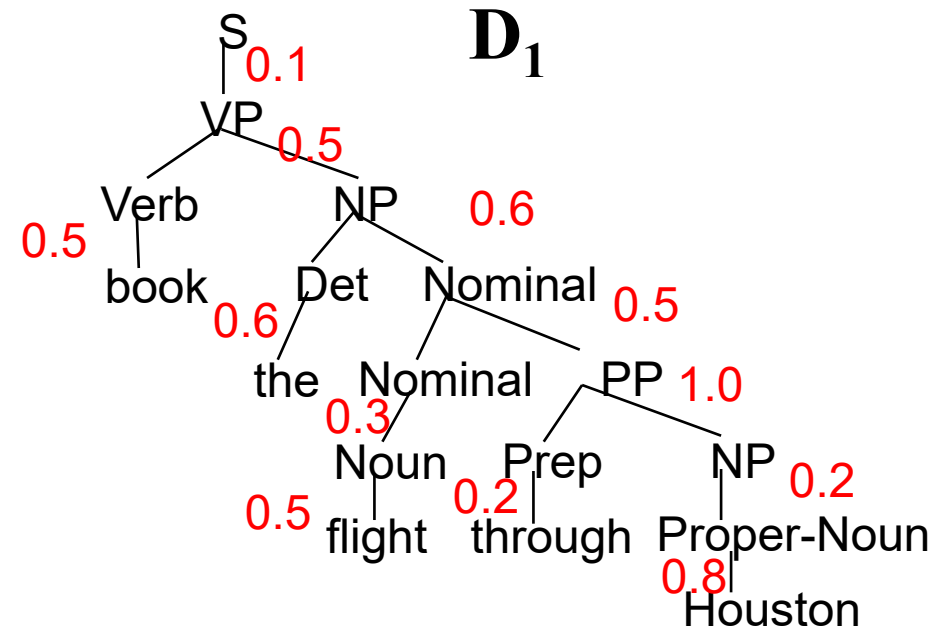
Lexicon

Det $\rightarrow$ the | a | that | this
 0.6 0.2 0.1 0.1
 Noun $\rightarrow$ book | flight | meal | money
 0.1 0.5 0.2 0.2
 Verb $\rightarrow$ book | include | prefer
 0.5 0.2 0.3
 Pronoun $\rightarrow$ I | he | she | me
 0.5 0.1 0.1 0.3
 Proper-Noun $\rightarrow$ Houston | NWA
 0.8 0.2
 Aux $\rightarrow$ does
 1.0
 Prep $\rightarrow$ from | to | on | near | through
 0.25 0.25 0.1 0.2 0.2

Sentence Probability

- Probability of derivation is the product of the probabilities of its productions.

$$\begin{aligned}P(D_1) &= 0.1 \times 0.5 \times 0.5 \times 0.6 \times 0.6 \times \\ &\quad 0.5 \times 0.3 \times 1.0 \times 0.2 \times 0.2 \times \\ &\quad 0.5 \times 0.8 \\ &= 0.0000216\end{aligned}$$



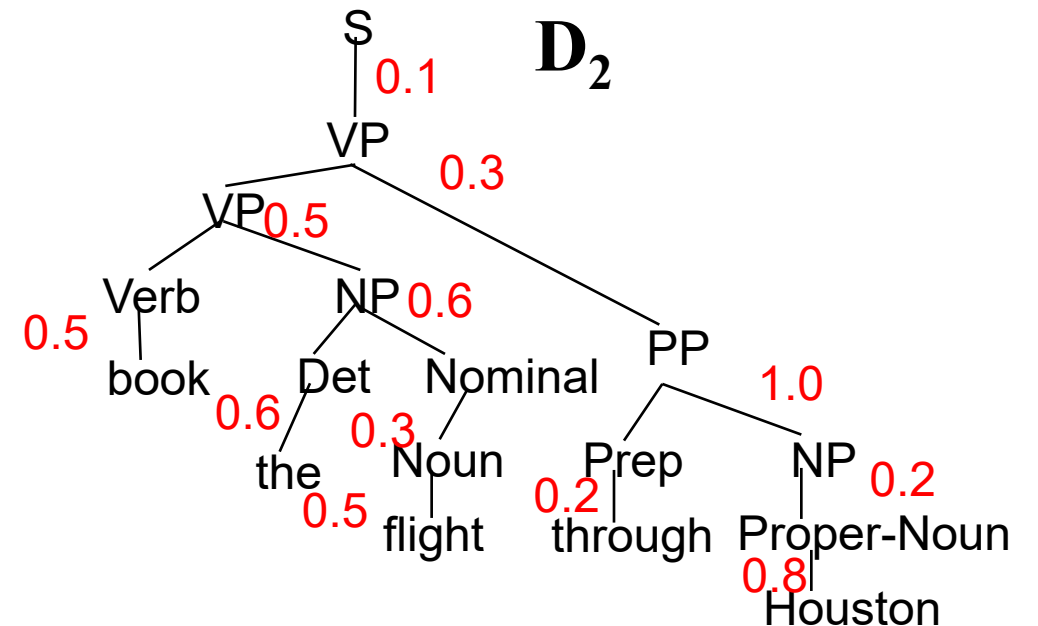
Syntactic Disambiguation

- Resolve ambiguity by picking most probable parse tree.

$$\begin{aligned} P(D_2) &= 0.1 \times 0.3 \times 0.5 \times 0.6 \times 0.5 \times \\ &\quad 0.6 \times 0.3 \times 1.0 \times 0.5 \times 0.2 \times \\ &\quad 0.2 \times 0.8 \\ &= 0.00001296 \end{aligned}$$

$$P(D_1) > P(D_2)$$

Tree D1 is more probable!



Sentence Probability

- Probability of a sentence is the sum of the probabilities of all of its derivations.

$$\begin{aligned} P(\text{"book the flight through Houston"}) &= \\ P(D_1) + P(D_2) &= 0.0000216 + 0.00001296 \\ &= 0.00003456 \end{aligned}$$

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Probabilistic CKY

- CKY can be modified for PCFG parsing by including in each cell a probability for each non-terminal.
- When transforming the grammar to CNF, must set production probabilities to preserve the probability of derivations.

Probabilistic Grammar Conversion

Original Grammar

$S \rightarrow NP VP$ 0.8

$S \rightarrow Aux NP VP$ 0.1

$S \rightarrow VP$ 0.1

$NP \rightarrow Pronoun$ 0.2

$NP \rightarrow Proper-Noun$ 0.2

$NP \rightarrow Det Nominal$ 0.6

$Nominal \rightarrow Noun$ 0.3

$Nominal \rightarrow Nominal Noun$ 0.2

$Nominal \rightarrow Nominal PP$ 0.5

$VP \rightarrow Verb$ 0.2

$VP \rightarrow Verb NP$ 0.5

$VP \rightarrow VP PP$ 0.3

$PP \rightarrow Prep NP$ 1.0

Chomsky Normal Form

$S \rightarrow NP VP$ 0.8

$S \rightarrow X1 VP$ 0.1

$X1 \rightarrow Aux NP$ 1.0

$S \rightarrow book \mid include \mid prefer$
0.01 0.004 0.006

$S \rightarrow Verb NP$ 0.05

$S \rightarrow VP PP$ 0.03

$NP \rightarrow I \mid he \mid she \mid me$
0.1 0.02 0.02 0.06

$NP \rightarrow Houston \mid NWA$
0.16 .04

$NP \rightarrow Det Nominal$ 0.6

$Nominal \rightarrow book \mid flight \mid meal \mid money$
0.03 0.15 0.06 0.06

$Nominal \rightarrow Nominal Noun$ 0.2

$Nominal \rightarrow Nominal PP$ 0.5

$VP \rightarrow book \mid include \mid prefer$
0.1 0.04 0.06

$VP \rightarrow Verb NP$ 0.5

$VP \rightarrow VP PP$ 0.3

$PP \rightarrow Prep NP$ 1.0

Adding **dummy non-terminals**, and **setting the probability** of the new rule with the dummy nonterminal on the LHS **to 1**

Normalise to preserve the original probabilities

Probabilistic CKY

Length = 1

	Book j=1	the 2	flight 3	through 4	Houston 5
i=0	S :.01, VP:.1, Verb:.5 Nominal:.03 Noun:.1				
1		Det:.6			
2			Nominal:.15 Noun:.5		
3				Prep:.2	
4					NP:.16 PropNoun:.8

R1: S → NP VP 0.8
 R2: S → X1 VP 0.1
 R3: X1 → Aux NP 1.0
 S → **book** | include | prefer
 0.01 0.004 0.006
 R4: S → Verb NP 0.05
 R5: S → VP PP 0.03
 NP → I | he | she | me
 0.1 0.02 0.02 0.06
 NP → **Houston** | NWA
 0.16 .04
 R6: NP → Det Nominal 0.6
 Nominal → **book** | **flight** | meal | money
 0.03 0.15 0.06 0.06
 R7: Nominal → Nominal Noun 0.2
 R8: Nominal → Nominal PP 0.5
 VP → **book** | include | prefer
 0.1 0.04 0.06
 R9: VP → Verb NP 0.5
 R10: VP → VP PP 0.3
 R11: PP → Prep NP 1.0
 Det → **the** | a | that | this
 0.6 0.2 0.1 0.1
 Prep → from | to | on | near | **through**
 0.25 0.25 0.1 0.2 0.2

Probabilistic CKY

Length = 2

	Book j=1	the 2	flight 3	through 4	Houston 5
i=0	S :.01, VP:.1, Verb:.5 Nominal:.03 Noun:.1	None			
1		Det:.6	NP (R6, 1): 6*.6*.15 =.054		
2			Nominal:.15 Noun:.5	None	
3				Prep:.2	PP (R11, 4): 1.0*.2*.16 =.032
4					NP:.16 PropNoun:.8

R1: S → NP VP 0.8
 R2: S → X1 VP 0.1
 R3: X1 → Aux NP 1.0
 S → **book** | include | prefer
 0.01 0.004 0.006
 R4: S → Verb NP 0.05
 R5: S → VP PP 0.03
 NP → I | he | she | me
 0.1 0.02 0.02 0.06
 NP → **Houston** | NWA
 0.16 .04
R6: NP → Det Nominal 0.6
 Nominal → **book** | **flight** | meal | money
 0.03 0.15 0.06 0.06
 R7: Nominal → Nominal Noun 0.2
 R8: Nominal → Nominal PP 0.5
 VP → **book** | include | prefer
 0.1 0.04 0.06
 R9: VP → Verb NP 0.5
 R10: VP → VP PP 0.3
R11: PP → Prep NP 1.0
 Det → **the** | a | that | this
 0.6 0.2 0.1 0.1
 Prep → from | to | on | near | **through**
 0.25 0.25 0.1 0.2 0.2

Probabilistic CKY

Length = 3

	Book j=1	the 2	flight 3	through 4	Houston 5
i=0	S :.01, VP:.1 Verb:.5 Nominal:.03 Noun:.1	None	VP (R9, 1): .5*.5*.054 =.0135 S (R4, 1): 05*.5*.054 =.00135		
1		Det:.6	NP (R6, 1): 6*.6*.15 =.054	None	
2			Nominal:.15 Noun:.5	None	Nominal (R8, 3): .5*.15*.032 =.0024
3				Prep:.2	PP (R11, 4): 1.0*.2*.16 =.032
4					NP:.16 PropNoun:.8

R1: S → NP VP 0.8

R2: S → X1 VP 0.1

R3: X1 → Aux NP 1.0

S → **book** | include | prefer
0.01 0.004 0.006

R4: S → Verb NP 0.05

R5: S → VP PP 0.03

NP → I | he | she | me
0.1 0.02 0.02 0.06

NP → **Houston** | NWA
0.16 .04

R6: NP → Det Nominal 0.6

Nominal → **book** | **flight** | meal | money
0.03 0.15 0.06 0.06

R7: Nominal → Nominal Noun 0.2

R8: Nominal → Nominal PP 0.5

VP → **book** | include | prefer
0.1 0.04 0.06

R9: VP → Verb NP 0.5

R10: VP → VP PP 0.3

R11: PP → Prep NP 1.0

Det → **the** | a | that | this
0.6 0.2 0.1 0.1

Prep → from | to | on | near | **through**
0.25 0.25 0.1 0.2 0.2

Probabilistic CKY

Length = 4

	Book	the	flight	through	Houston
	j= 1	2	3	4	5
i= 0	S :.01, VP:.1, Verb:.5 Nominal:.03 Noun:.1	None	VP (R9, 1): .0135 S (R4, 1): 00135	None	
1		Det:.6	NP (R6, 1): 6*.6*.15 =.054	None	NP (R6, 2): 6*.6*.0024 =.000864
2			Nominal:.15 Noun:.5	None	Nominal (R8, 3): .5*.15*.032 =.0024
3				Prep:.2	PP (R11, 4): 1.0*.2*.16 =.032
4					NP:.16 PropNoun:.8

- R1: S → NP VP 0.8
R2: S → X1 VP 0.1
R3: X1 → Aux NP 1.0
S → **book** | include | prefer
0.01 0.004 0.006
R4: S → Verb NP 0.05
R5: S → VP PP 0.03
NP → I | he | she | me
0.1 0.02 0.02 0.06
NP → **Houston** | NWA
0.16 .04
R6: NP → Det Nominal 0.6
Nominal → **book** | **flight** | meal | money
0.03 0.15 0.06 0.06
R7: Nominal → Nominal Noun 0.2
R8: Nominal → Nominal PP 0.5
VP → **book** | include | prefer
0.1 0.04 0.06
R9: VP → Verb NP 0.5
R10: VP → VP PP 0.3
R11: PP → Prep NP 1.0
Det → **the** | a | that | this
0.6 0.2 0.1 0.1
Prep → from | to | on | near | **through**
0.25 0.25 0.1 0.2 0.2

Probabilistic CKY

Length = 5

	Book j= 1	the 2	flight 3	through 4	Houston 5
i= 0	S :.01, VP:.1, Verb:.5 Nominal:.03 Noun:.1	None	VP (R9, 1): .0135 S (R4, 1): 00135	None	S (R4, 1): .0000216 S (R5, 3): .00001296
1		Det:.6	NP (R6, 1): 6*.6*.15 =.054	None	NP (R6, 2): 6*.6*.0024 =.000864
2			Nominal:.15 Noun:.5	None	Nominal (R8, 3): .5*.15*.032 =.0024
3				Prep:.2	PP (R11, 4): 1.0*.2*.16 =.032
4					NP:.16 PropNoun:.8

R1: S → NP VP 0.8
R2: S → X1 VP 0.1
R3: X1 → Aux NP 1.0
S → **book** | include | prefer
0.01 0.004 0.006
R4: S → Verb NP 0.05
R5: S → VP PP 0.03
NP → I | he | she | me
0.1 0.02 0.02 0.06
NP → **Houston** | NWA
0.16 .04
R6: NP → Det Nominal 0.6
Nominal → **book** | **flight** | meal | money
0.03 0.15 0.06 0.06
R7: Nominal → Nominal Noun 0.2
R8: Nominal → Nominal PP 0.5
VP → **book** | include | prefer
0.1 0.04 0.06
R9: VP → Verb NP 0.5
R10: VP → VP PP 0.3
R11: PP → Prep NP 1.0
Det → **the** | a | that | this
0.6 0.2 0.1 0.1
Prep → from | to | on | near | **through**
0.25 0.25 0.1 0.2 0.2

Probabilistic CKY

Length = 5

	Book j= 1	the 2	flight 3	through 4	Houston 5
i= 0	S :.01, VP:.1 Verb:.5 Nominal:.03 Noun:.1	None	VP (R9, 1): .0135 S (R4, 1): 00135	None	S (R4, 1): .0000216 S (R5, 3): .00001296
1		Det:.6	NP (R6, 1): 6*.6*.15 =.054	None	NP (R6, 2): 6*.6*.0024 =.000864
2			Nominal:.15 Noun:.5	None	Nominal (R8, 3): .5*.15*.032 =.0024
3				Prep:.2	PP (R11, 4): 1.0*.2*.16 =.032
4					NP:.16 PropNoun:.8

Pick most probable
parse, i.e. take max.

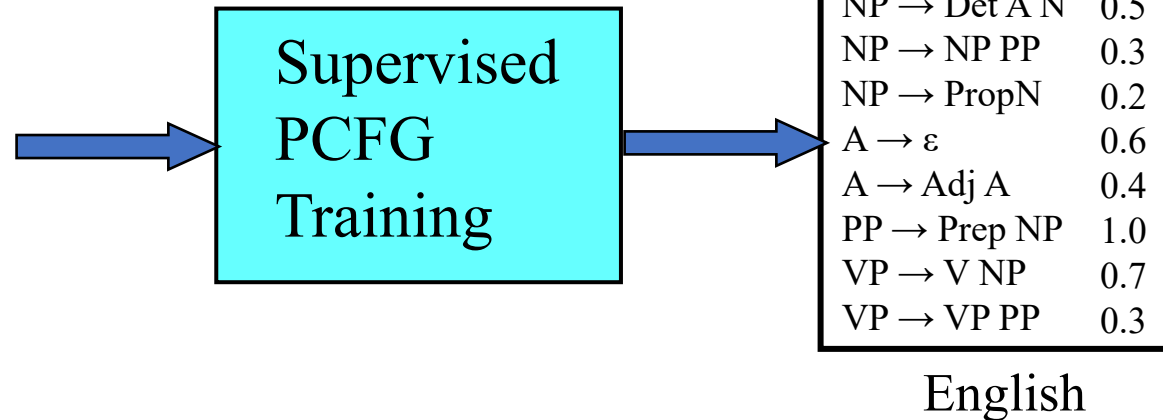
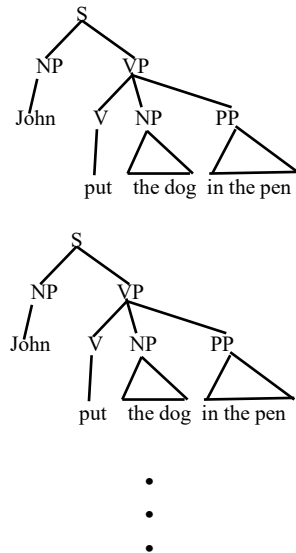
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PCFG: Supervised Training

- If parse trees are provided for training sentences, a grammar and its parameters can all be estimated directly from counts accumulated from the **tree-bank** (with appropriate smoothing).

Tree Bank



Trebanks

- **English Penn Treebank**: Standard corpus for testing syntactic parsing consists of 1.2 M words of text from the Wall Street Journal (WSJ).
- Typical to train on about 40,000 parsed sentences and test on an additional standard disjoint test set of 2,416 sentences.
- **Chinese Penn Treebank**: 100K words from the Xinhua news service.
- Other corpora existing in many languages, see the Wikipedia article “Treebank”
- See also <http://universaldependencies.org/>

First WSJ Sentence

```
( (S
  (NP-SBJ
    (NP (NNP Pierre) (NNP Vinken) )
    ( , , )
    (ADJP
      (NP (CD 61) (NNS years) )
      (JJ old) )
    ( , , ) )
  (VP (MD will)
    (VP (VB join)
      (NP (DT the) (NN board) )
      (PP-CLR (IN as)
        (NP (DT a) (JJ nonexecutive) (NN director) ))
      (NP-TMP (NNP Nov.) (CD 29) )))
  ( . . ) ) )
```

Estimating Production Probabilities

- Set of production rules can be taken directly from the set of rewrites in the treebank.
- Parameters can be directly estimated from frequency counts in the treebank.

$$P(\alpha \rightarrow \beta \mid \alpha) = \frac{\text{count}(\alpha \rightarrow \beta)}{\sum_{\gamma} \text{count}(\alpha \rightarrow \gamma)} = \frac{\text{count}(\alpha \rightarrow \beta)}{\text{count}(\alpha)}$$

PCFG: Maximum Likelihood Training

- Given a set of sentences, induce a grammar that maximizes the probability that this data was generated from this grammar.
- Assume the number of non-terminals in the grammar is specified.

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Lexicalize PCFG

- PCFGs can't distinguish between “eat sushi with chopsticks” and “eat sushi with tuna”.

- We need to take words into account

$$P(\text{VP}_{\text{eat}} \rightarrow \text{VP PP}_{\text{with chopsticks}} \mid \text{VP}_{\text{eat}})$$

$$\text{vs. } P(\text{VP}_{\text{eat}} \rightarrow \text{VP PP}_{\text{with tuna}} \mid \text{VP}_{\text{eat}})$$

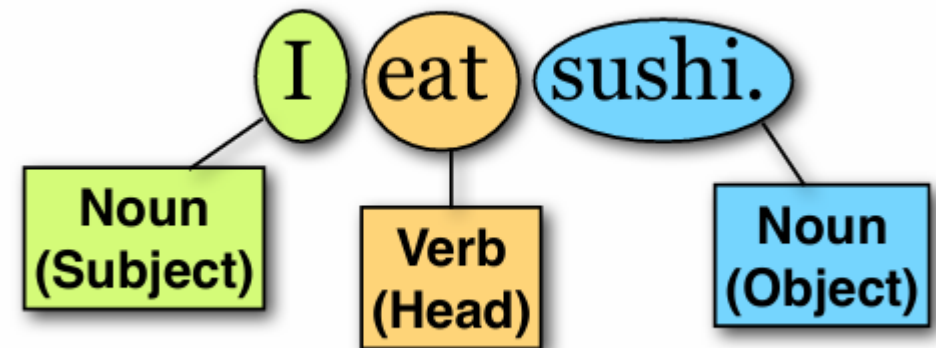
- Problem: sparse data

- $(\text{PP}_{\text{with fatty|whitel... tuna....}})$

- In order to work well, PCFGs must be **lexicalized**, i.e. productions must be specialized to specific words by including their **head-word** in their LHS non-terminals (e.g. VP-eat).

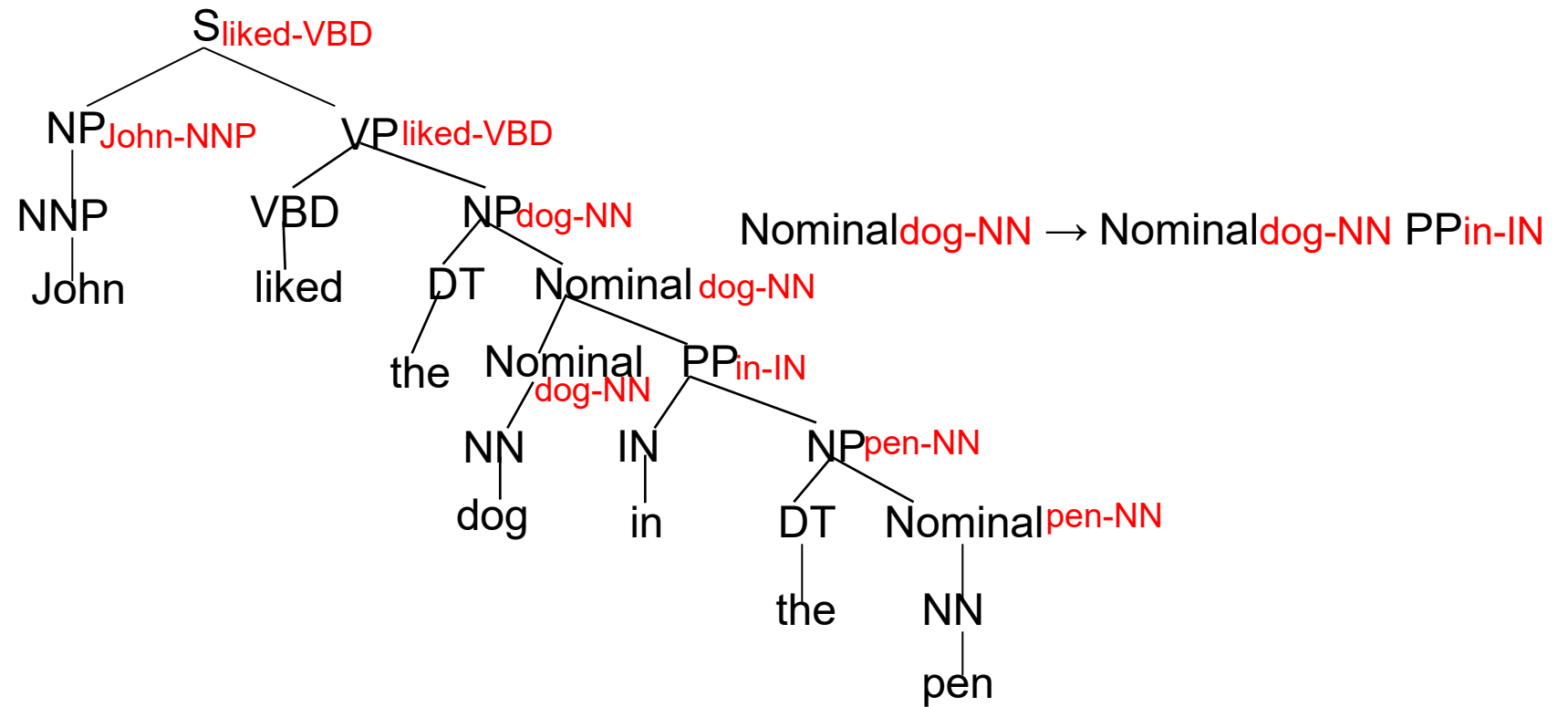
Head Words

- Syntactic phrases usually have a word in them that is most “central” to the phrase.
- Simple rules can identify the head of any phrase by percolating head words up the parse tree.
 - Head of a VP is the main verb
 - Head of an NP is the main noun
 - Head of a PP is the preposition
 - Head of a sentence is the head of its VP



Lexicalization Productions

- Specialized productions can be generated by including the head word and its POS of each non-terminal as part of that non-terminal's symbol.



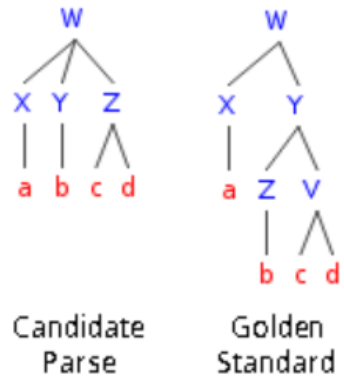
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Parsing Evaluation Metrics

- Measure the fraction of the **constituents** that match between the computed and human parse trees. If P is the system's parse tree and T is the human parse tree (the “gold standard”):
 - **Recall** = $(\# \text{ correct constituents in } P) / (\# \text{ constituents in } T)$
 - **Precision** = $(\# \text{ correct constituents in } P) / (\# \text{ constituents in } P)$
- **Labeled Precision** and **labeled recall** require getting the non-terminal label on the constituent node correct to count as correct.
- **F_1** is the harmonic mean of precision and recall.

Evaluate Example



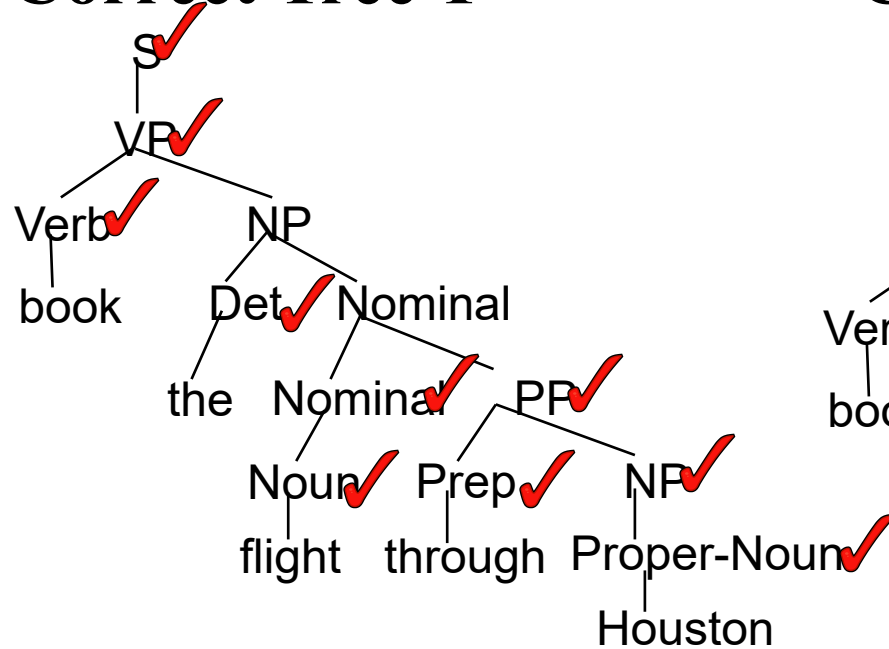
The constituents for each are given below in form label:yield (**yield** is the span of the node/constituent).

Candidate	gold
X:a	X:a
Y:b	Z:b
Z:cd	V:cd
--	Y:bcd
W:abcd	W:abcd

<https://linguistics.stackexchange.com/questions/1873/is-there-a-well-established-metric-to-measure-the-effectiveness-of-a-parsing-alg>

Evaluation Example

Correct Tree T

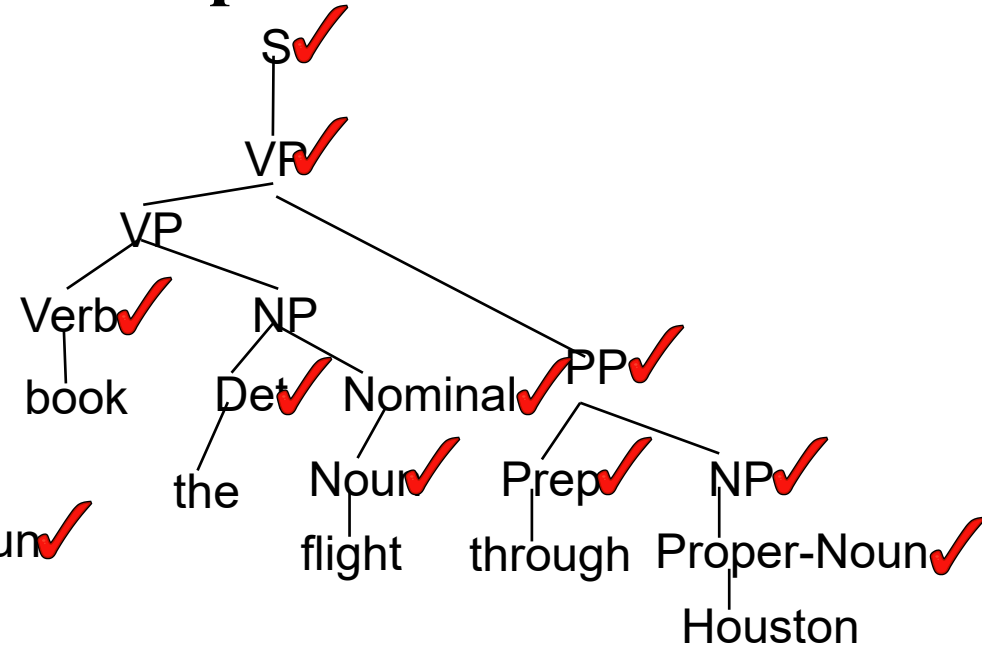


Constituents: 12

Correct Constituents: 10

Recall = $10/12 = 83.3\%$ Precision = $10/12 = 83.3\%$ $F_1 = 83.3\%$

Computed Tree P



Constituents: 12

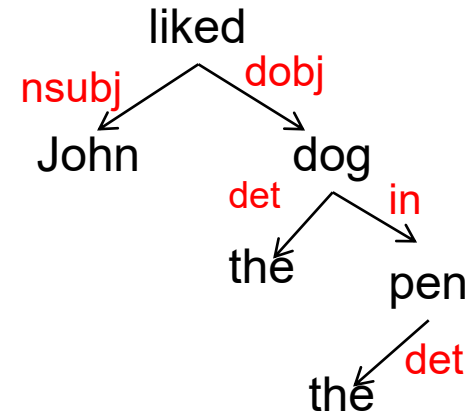
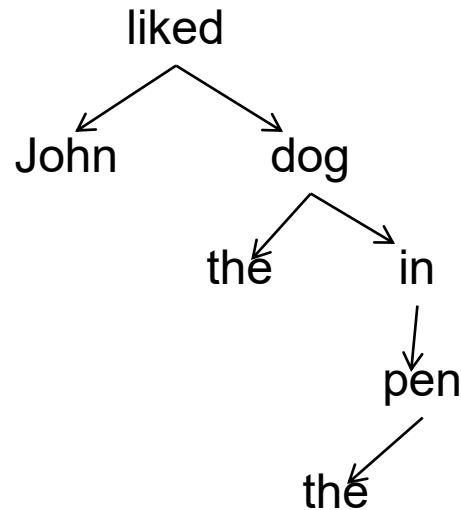
Note: It is only marked as correct if it covers the same span of input (i.e., look at the 2 Nominals in T and the single Nominal in P).

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Dependency Grammars

- An alternative to phrase-structure grammar is to define a parse as a directed graph between the words of a sentence representing *dependencies* between the words.



There are other grammar formalism too: [TAG](#), [CCG](#), [TSG](#).

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CKY Parsing... Some Challenges

- $O(n^3)$ complexity
- Grammar must be converted to **Chomsky Normal Form**, all rules of form $A \rightarrow BC$ or $A \rightarrow w$. Distorts natural linguistic structure.
- Vanilla CKY treats all matching rules equally. No mechanism to prefer "I saw the man with the telescope" over its **ambiguous parse**.
- **Rules** must come from somewhere, **hand-crafted grammars** don't scale; learning from treebanks introduces coverage gaps.
- Words treated as **discrete tokens**; no access to embeddings or contextual representations from pre-training

Advancements

Parsing with Compositional Vector Grammars

Richard Socher John Bauer Christopher D. Manning Andrew Y. Ng

Computer Science Department, Stanford University, Stanford, CA 94305, USA

richard@socher.org, horatio@gmail.com, manning@stanford.edu, ang@cs.stanford.edu

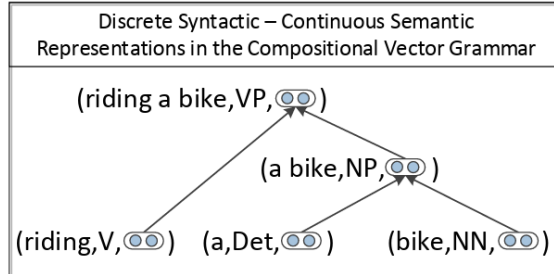


Figure 1: Example of a CVG tree with (category,vector) representations at each node. The vectors for nonterminals are computed via a new type of recursive neural network which is conditioned on syntactic categories from a PCFG.

- Traditional PCFGs and n-grams struggle with unseen or long phrases and cannot capture meaning well.
- **Core idea (CVG):** Jointly learn **syntax (tree structure)** and **semantic vector representations** of phrases.
- Combine **PCFG (discrete grammar rules)** with **neural composition (continuous vectors)** to resolve meaning-based ambiguities.
- Use **syntactically untied weights**, so different phrase types are combined differently, leading to better parsing accuracy

ACL, 2013

Grammar as a Foreign Language

Oriol Vinyals*

Google

vinyals@google.com

Lukasz Kaiser*

Google

lukaszkaizer@google.com

Terry Koo

Google

terrykoo@google.com

Slav Petrov

Google

slav@google.com

Ilya Sutskever

Google

ilyasu@google.com

Geoffrey Hinton

Google

geoffhinton@google.com

“ We cast the problem of constituency parsing as a sequence-to-sequence mapping problem... Our model can be viewed as a sequence-to-sequence model that reads words of a sentence and outputs its parse tree encoded as a sequence of bracketed labels.”

NeurIPS, 2015

- **LSTM encoder** reads the input sentence and produces a fixed context vector. An **LSTM decoder with attention** generates the linearized bracket string token by token (each open-bracket, label, word, close-bracket is a separate output token).
- **No grammar rules, no CKY.** The model learns valid tree structure purely from (sentence, parse) pairs in PTB training data. Beam search used at decode time.

Approximating CKY with Transformers

EMNLP, 2023

Ghazal Khalighinejad

Duke University
gk126@duke.edu

Ollie Liu

University of Southern California
zliu2898@usc.edu

Sam Wiseman

Duke University
swiseman@cs.duke.edu

Constituency Parsing using LLMs

Xuefeng Bai[✉], Jialong Wu[✉], Yulong Chen[✉], Zhongqing Wang[✉],
Kehai Chen[✉], Min Zhang[✉], Yue Zhang[✉], *Senior Member, IEEE*

IEEE Transactions on Audio, Speech and Language Processing (2025)

Task Introduction

You will be given one sentence for constituency parsing. Constituency parsing is the process of analyzing the sentences by breaking down it into sub-phrases also known as constituents. Here are some bracket constituents for you to learn from.

Clause Level
S – simple declarative clause
...

Phrase Level
ADJP – Adjective Phrase
...

Word level
CC – Coordinating conjunction
...

Instruction

Generate the constituency tree of the given sentence. You should output the constituency tree as a bracket sequence which uses parentheses to mark visit depth and restore tree structure.

Demonstrations (Few-shot only)

The constituent tree of sentence "Singapore is located in Asia" is:
(S (NP (NNP Singapore)) (VP (VBZ is) (VP (VBN located) (PP (IN in) (NP (NNP Asia))))))
...

Task input

The constituent tree of sentence "Sam thinks Sandy Likes the book" is:

Fig. 3. Illustration of prompts for zero/few-shot learning.

Feedback time!



** Please submit weekly Feedback! **

